Predicting Thermophysical Properties of Organic Compounds and Aviation Fuel Mixtures from Nuclear Magnetic Resonance Spectra

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Abstract

Rapid prediction of thermophysical properties from molecular spectra would accelerate the qualification of sustainable aviation fuels, a process that currently requires years of physical testing per candidate pathway. This report develops a four-phase machine learning pipeline that predicts temperature-dependent viscosity and surface tension directly from proton nuclear magnetic resonance (NMR) spectra. Phase 1 establishes baselines using Morgan fingerprints and RDKit descriptors, achieving cross-validated $R^2 = 0.64$ for viscosity and $R^2 = 0.56$ for surface tension on 856 compounds. Phase 2 replaces fingerprints with 120 domain-informed features extracted from experimental 1H NMR peak lists, demonstrating that combined NMR and fingerprint representations reach hydrocarbon $R^2 = 0.69$ for viscosity and $R^2 = 0.81$ for surface tension, though NMR alone lacks molecular weight information critical for viscosity. Phase 3 extends prediction to real fuel mixtures, validating against four AFRL reference fuels (JP-5, Jet-A, HRJ-Camelina, Shell SPK, Sasol IPK, Gevo ATJ) at a baseline of 53.8% mean absolute percentage error (MAPE). Phase 4 introduces transfer learning: a multilayer perceptron pretrained on 296,000 NMR-to-descriptor mappings from the NMRexp database is fine-tuned on 943 compounds at the NMR–viscosity intersection, reducing fuel viscosity MAPE to 29.3%, a 46% relative improvement. The pipeline draws on fourteen NMR spectral databases totaling 1.8 million compounds, six thermophysical property sources exceeding one million data points, and six fuel-specific datasets. Systematic experiments with Arrhenius-structured networks, embedding extraction, cold-temperature augmentation, sample weighting, and synthetic NMR generation all failed to improve beyond the fine-tuned transfer learning result, identifying data quality at the NMR–viscosity intersection, not model architecture, as the binding constraint. All code, data pipelines, and trained model weights are publicly available.

1. Introduction

The prediction of thermophysical properties from molecular structure is a central problem in chemical engineering, combustion science, and fuel qualification. For decades, empirical correlations and group contribution methods have served as the primary tools for estimating properties such as viscosity, surface tension, and density from knowledge of a compound's functional groups [1], [2], [3]. These approaches work well for simple molecules within their training domain, but they struggle with the structural diversity and compositional complexity of real-world fuel mixtures. Machine learning offers a path beyond these limitations, provided that the molecular representation fed to the model captures enough structural information to support accurate generalization. This report investigates nuclear magnetic resonance spectroscopy as precisely such a representation. An NMR spectrum records, in a single rapid measurement, the complete distribution of hydrogen and carbon environments within a molecule or mixture, encoding the functional group inventory, connectivity patterns, and relative abundances that govern bulk thermophysical behavior. The central hypothesis is that NMR spectral features, when combined with modern regression methods, can predict temperature-dependent viscosity and surface tension with accuracy sufficient to accelerate the screening and qualification of aviation fuels.

The remainder of this introduction is organized as follows. Section 1.1 provides a self-contained overview of NMR spectroscopy for readers who may not have a chemistry or spectroscopy background, establishing the physical principles and spectral features that underpin the molecular descriptors used throughout this work. Section 1.2 motivates the practical application by describing the thermophysical properties critical to aviation fuel performance and the regulatory bottleneck that sustainable aviation fuel (SAF) candidates face during qualification. Section 1.3 summarizes the four-phase modeling pipeline developed in this report and highlights the principal contributions.

1.1. Nuclear Magnetic Resonance Spectroscopy

Nuclear magnetic resonance spectroscopy exploits a quantum-mechanical property of atomic nuclei known as spin. Certain isotopes, including hydrogen-1 (^{1}H) and carbon-13 (^{13}C), possess a nonzero spin angular momentum that causes them to behave as small magnetic dipoles. When an external magnetic field B_0 is applied, these nuclear dipoles align either with or against the field, occupying discrete energy levels separated by a gap $\Delta E = \gamma \hbar B_0$, where γ is the gyromagnetic ratio specific to each isotope and $\hbar$ is the reduced Planck constant. Irradiating the sample with radiofrequency energy that matches this energy gap induces transitions between spin states, a condition called resonance. The frequencies at which different nuclei resonate encode detailed information about their electronic environment and, by extension, the molecular structure surrounding them [4], [5]. Modern NMR spectrometers detect the free induction decay after a broadband excitation pulse and recover the frequency-domain spectrum by Fourier transformation. The entire measurement typically requires fewer than five minutes and consumes only microliters of sample, making NMR one of the most efficient analytical techniques available for organic structure determination.

Proton (^{1}H) NMR is the most widely practiced variant of the experiment for two complementary reasons. First, the hydrogen-1 nucleus has the largest gyromagnetic ratio of any stable isotope other than tritium, giving it the highest intrinsic sensitivity per nucleus. Second, hydrogen is the most abundant element in organic molecules and is present in nearly every functional group of interest, so a single ^{1}H spectrum surveys the entire carbon skeleton indirectly through its attached and nearby protons [6], [7]. These advantages make ^{1}H NMR the default analytical tool for structure elucidation in organic chemistry, pharmaceutical development, and petrochemical analysis. Carbon-13 NMR provides complementary information by directly observing the carbon framework, but its natural abundance is only 1.1%, resulting in sensitivity roughly 5,700 times lower than that of ^{1}H. In practice, ^{13}C spectra require longer acquisition times or isotopic enrichment and are therefore used as a supplement to, rather than a replacement for, proton NMR [4].

The horizontal axis of a ^{1}H NMR spectrum is reported in parts per million (ppm) of chemical shift δ rather than in absolute frequency units. Chemical shift arises because the electrons surrounding each proton generate small local magnetic fields that partially shield the nucleus from the full external field B_0. Protons in electron-

rich environments experience greater shielding and resonate at lower frequencies (smaller δ), while those near electronegative atoms or aromatic ring currents experience reduced shielding and resonate at higher frequencies (larger δ). All chemical shifts are measured relative to tetramethylsilane (TMS), which is assigned $\delta = 0$ ppm by convention. Expressing the spectrum in ppm rather than hertz makes the peak positions independent of the spectrometer's operating frequency, so that spectra acquired on instruments of different field strengths can be compared directly [5]. In practice, the ^{1}H chemical shift range spans roughly 0 to 12 ppm. Aliphatic methyl and methylene groups appear between 0.5 and 2.0 ppm, protons adjacent to oxygen or nitrogen functionalities fall near 2.0 to 4.5 ppm, olefinic protons resonate between 4.5 and 6.5 ppm, aromatic ring protons occupy the 6.5 to 8.5 ppm region, and aldehyde or carboxylic acid protons appear furthest downfield near 9 to 12 ppm [6]. Because each class of chemical environment maps onto a characteristic ppm window, even a cursory visual inspection of the spectrum reveals which functional groups are present and in what relative proportions.

Beyond the position of each signal, the fine structure of NMR peaks carries additional structural information through a phenomenon known as scalar coupling (also called J-coupling or spin-spin splitting). When a proton has n nonequivalent neighboring protons separated by two or three chemical bonds, its NMR signal splits into $n + 1$ lines. A proton with no coupled neighbors appears as a singlet, one neighboring proton produces a doublet, two neighbors produce a triplet, and so on. The spacing between the lines of a multiplet, measured in hertz and called the coupling constant J, is independent of the external magnetic field strength and depends instead on the dihedral angle, bond order, and hybridization connecting the coupled nuclei. Typical vicinal ($^3J_{HH}$) coupling constants range from about 2 Hz in strained rings to 16 Hz across trans-disubstituted double bonds [4]. Because coupling constants encode geometric relationships between bonded atoms, they provide information about molecular conformation that chemical shift alone cannot supply. Two-dimensional NMR experiments such as COSY (correlation spectroscopy), which maps ^{1}H–^{1}H coupling networks, and HSQC (heteronuclear single quantum coherence), which correlates each proton with its directly bonded ^{13}C, extend this connectivity analysis further, although the one-dimensional ^{1}H spectrum remains the primary data source for the machine-learning models developed in this work [8].

A third dimension of the ^{1}H spectrum is integration. The area under each signal is directly proportional to the number of chemically equivalent protons giving rise to that signal, a property that is unique to NMR among common spectroscopic methods. A peak integrating as 3 relative to a peak integrating as 1, for example, indicates a methyl group versus a lone proton. Taken together, chemical shift, multiplicity, coupling constants, and integration allow a trained analyst to reconstruct the full connectivity of an unknown organic molecule from its ^{1}H NMR spectrum alone, often without recourse to any other analytical technique [5].

These same attributes that make ^{1}H NMR indispensable for human analysts also make it well suited as input to machine-learning models. A spectrum can be represented as a discrete list of peaks, each described by a chemical shift value, a relative intensity, and a multiplicity label, yielding a compact, standardized numerical descriptor that varies smoothly with molecular structure. Unlike molecular graph representations that require specialized graph neural network architectures [9], [10], an NMR peak list or binned spectrum can be consumed directly by conventional regression, ensemble, and feed-forward neural network methods. NMR spectra are also among the most reproducible analytical measurements in chemistry. Standardized reference databases, including the Human Metabolome Database (HMDB) [11], NMRShiftDB2 [12], and NMRexp [13], store hundreds of thousands of quality-controlled spectra in uniform formats, providing a ready-made corpus for data-driven modeling. The Spectral Database for Organic Compounds (SDBS), maintained by the National Institute of Advanced Industrial Science and Technology in Japan, contributes an additional 35,000 ^{1}H spectra with verified assignments [14], [15]. The rich structural encoding inherent in ^{1}H NMR, combined with its experimental standardization and broad chemical coverage, motivates its use as the primary molecular descriptor throughout this work.

For fuel mixtures specifically, NMR offers an advantage that no single-compound technique can match. A ^{1}H spectrum of a multi-component fuel is the mole-fraction-weighted superposition of the spectra of its individual constituents [16], [17]. The integrated areas of characteristic spectral regions, such as the aromatic (6.5–8.5 ppm), olefinic (4.5–6.5 ppm), and various aliphatic windows, directly quantify the functional group inventory of the entire mixture without requiring prior separation or identification of individual species. Abdul Jameel and coworkers demonstrated that these NMR-derived functional group fractions, termed atom types, can predict ignition quality, octane number, and other combustion-relevant properties of complex fuel blends with accuracy comparable to

detailed compositional analysis [18], [19], [20]. Parker et al. extended this framework to density and surface tension predictions for aviation fuels using NMR atom type descriptors [21]. These precedents establish NMR as a proven front-end measurement for fuel property estimation and provide the methodological foundation on which the present work builds.

1.2. Thermophysical Properties and Aviation Fuel Qualification

The global aviation industry consumes more than 100 billion gallons of jet fuel annually, making it one of the largest single-commodity supply chains in the world [22]. Every gallon must satisfy stringent thermophysical specifications before it is cleared for use in turbine engines, because the consequences of off-specification fuel range from degraded engine performance to catastrophic in-flight failure. The governing standards, ASTM D1655 for conventional petroleum-derived jet fuel and ASTM D7566 for fuels containing synthesized hydrocarbons, impose limits on dozens of measurable properties [23], [24]. Among these, three properties exert an outsized influence on combustion system design and flight safety, and understanding them in some detail is essential to appreciating why accurate prediction matters.

Viscosity, the resistance of a fluid to shear flow, governs two aspects of fuel system performance that are critical under extreme conditions. At the low temperatures encountered during high-altitude cruise (below $-40°C$ for many flight profiles), excessive viscosity impedes fuel flow through pumps, filters, and transfer lines, potentially starving the engine of fuel. At the high temperatures found in fuel-return lines near the combustor (above $100°C$), viscosity determines the quality of fuel atomization at the injector tip: a fuel that is too thin produces a spray of excessively fine droplets that evaporate prematurely, while one that is too thick yields large droplets that penetrate too far into the combustion chamber and burn incompletely [25]. ASTM D1655 sets an upper viscosity limit of 8.0 $\mathrm{mm}^2/\,\mathrm{s}$ at $-20°C$ for Jet A fuel and $-40°C$ for Jet A-1, ensuring adequate pumpability at the coldest anticipated service temperatures [23]. The temperature dependence of liquid viscosity is well described by an Arrhenius-type relationship,

$$\ln \eta = A + \frac{B}{T} \tag{1}$$

where η is the dynamic viscosity, T is absolute temperature, and A and B are substance-specific constants [26]. This exponential sensitivity means that small compositional changes in a fuel blend can shift the viscosity-temperature curve by enough to violate specification limits, making accurate prediction across the full temperature range essential for fuel qualification.

Surface tension, the energy per unit area at the liquid-air interface, controls the initial breakup of the fuel jet emerging from the injector nozzle and the subsequent formation, coalescence, and evaporation of droplets within the combustion zone. Lower surface tension promotes finer atomization and faster evaporation, generally improving combustion efficiency and reducing soot formation. Higher surface tension produces larger droplets that require more time and distance to vaporize, shifting the heat-release profile downstream and increasing the risk of hot spots and incomplete combustion [21]. Although surface tension varies less dramatically with temperature than viscosity does (typically decreasing linearly at a rate of roughly 0.1 mN / m per degree Celsius for hydrocarbons), its influence on spray dynamics makes it a key parameter in combustor design models. Density, the third property of primary interest, determines the volumetric energy content of the fuel and therefore the range achievable for a given tank volume. Fuel density also affects the momentum of the fuel jet and the pressure drop through metering orifices, both of which influence injector performance [22].

Traditional measurement of these properties is slow and sample-intensive, compounding the bottleneck in fuel qualification. Viscosity determination by capillary viscometry following ASTM D445 requires several milliliters of purified sample, temperature-controlled baths equilibrated to within $\pm 0.01°C$, and 30 to 60 minutes per data point [27]. Surface tension measurement by the pendant-drop or Du Nouy ring method is similarly time-consuming. For a candidate fuel of unknown composition, a full property screening across the relevant temperature range, from $-40°C$ to $100°C$, can consume days of laboratory time and hundreds of milliliters of material that may have been produced only at bench scale. These constraints create a powerful incentive for predictive methods that can estimate fuel properties from minimal experimental input.

The urgency of rapid property prediction is amplified by the global push toward sustainable aviation fuels. Aviation accounts for roughly 2 to 3 percent of global carbon dioxide emissions, and decarbonizing the sector requires replacing petroleum-derived kerosene with fuels synthesized from biomass, waste oils, alcohols, or captured carbon dioxide [28]. Hundreds of candidate SAF pathways have been proposed, spanning Fischer-Tropsch synthesis, hydroprocessed esters and fatty acids, alcohol-to-jet conversion, and catalytic hydrothermolysis, among others. Yet only about ten of these pathways have received ASTM approval through the D4054 evaluation process [29]. The qualification of a single new fuel pathway typically requires three to five years of tiered evaluation, beginning with preliminary screening (Tier 1), proceeding through fit-for-purpose testing and rig-level combustion trials (Tiers 2 and 3), and culminating in full-scale engine tests (Tier 4), at a total cost that can reach tens of millions of dollars per pathway [25]. At that pace, the portfolio of approved SAFs will not grow fast enough to meet the aviation industry's net-zero-by-2050 commitments.

A predictive framework that can estimate key thermophysical properties from a rapid analytical measurement would compress the earliest stages of this pipeline. Rather than synthesizing and physically testing every candidate blend, researchers could acquire a single NMR spectrum of the fuel or feedstock, extract spectral descriptors, and use trained machine-learning models to predict viscosity, surface tension, and density across the full operating temperature range. Candidates whose predicted properties fall outside specification limits could be eliminated immediately, reserving expensive physical testing for the most promising formulations. The NMR measurement itself is non-destructive, requires no calibration standards, and provides quantitative composition data without prior knowledge of the analytes present, making it an ideal front-end for such a screening pipeline [16], [17]. This vision, from NMR spectrum to predicted fuel properties in minutes rather than months, motivates the modeling effort described in the remainder of this report.

1.3. Scope and Contributions

This report describes the development, training, and evaluation of a four-phase machine-learning pipeline that predicts temperature-dependent viscosity and surface tension from molecular and spectral descriptors. Each phase addresses a distinct modeling challenge, and the phases are designed to build on one another so that representations learned on large pure-compound datasets can be transferred to the data-scarce domain of real fuel mixtures.

Phase 1 establishes baseline models using molecular fingerprints and computed descriptors alone, without any NMR data. Morgan circular fingerprints (2048-bit) and 195 RDKit molecular descriptors are computed from SMILES strings for 856 organic compounds whose experimental viscosity and surface tension data are available in NIST ThermoML [30], [31] and the ChEDL chemicals library [32]. Several regression architectures are compared, including feed-forward neural networks, XGBoost [33], random forests [34], and ridge regression, under a rigorous compound-level cross-validation protocol that prevents temperature-based data leakage. Lasso feature selection reduces the descriptor space from nearly 1,700 dimensions to roughly 30 informative features per fold, and ensemble averaging over five independently trained models stabilizes predictions. Phase 1 serves two purposes: it quantifies the predictive ceiling achievable from molecular structure descriptors that do not require experimental measurement, and it provides the baseline against which NMR-based models are judged.

Phase 2 introduces NMR spectral features as molecular descriptors. Experimental ^{1}H NMR peak lists from NMRShiftDB2 [12] and NMRexp [13] are matched to the thermophysical property database by canonical SMILES, yielding a joint dataset of compounds for which both NMR spectra and measured properties are available. Three spectral encoding strategies are evaluated: binned histograms of the chemical shift axis, summary statistics of peak positions and intensities, and learned representations from a 1D convolutional neural network operating directly on the raw spectral vector. These NMR-only models are compared against the fingerprint baselines from Phase 1, and combined NMR-plus-fingerprint models are tested to determine whether NMR provides information that is complementary to computed molecular descriptors. The Phase 2 analysis reveals which spectral features carry the most predictive power and whether experimental NMR data can outperform purely computational molecular representations.

Phase 3 extends the pure-compound models to aviation fuel mixtures. Fuel-specific data, including NMR atom type distributions and measured bulk properties, are drawn from the Air Force Research Laboratory reference

fuel database [22], Parker et al. [21], and Abdul Jameel et al. [20]. Mixture-level descriptors are constructed by composition-weighting the pure-compound NMR features according to detailed hydrocarbon analysis or, where available, directly from the mixture-level ^{1}H spectrum. The models are validated against measured properties of JP-5, Jet A, and several alternative aviation fuels (Fischer-Tropsch, HEFA, and alcohol-to-jet blends), with prediction accuracy reported as mean absolute percentage error at each temperature point.

Phase 4 investigates whether pretraining on a large corpus of pure-compound NMR-property pairs improves prediction accuracy on the small fuel-mixture dataset. A multilayer perceptron is first trained to predict viscosity and surface tension for approximately 1.48 million pure-compound records, learning a general mapping from NMR descriptors to thermophysical behavior. The pretrained network's intermediate representations are then fine-tuned on the fuel mixture data from Phase 3. This transfer-learning strategy is motivated by the severe data scarcity in the fuel domain, where fewer than 30 fuels have comprehensive NMR and property characterization, and by the hypothesis that the relationship between NMR features and bulk properties is sufficiently transferable across chemical classes to benefit from large-scale pretraining.

The principal contributions of this work are as follows. First, it assembles the largest joint NMR-thermophysical-property dataset used for machine-learning property prediction to date, linking 3.37 million NMR spectra from NMRexp with over one million experimental property data points from NIST ThermoML. Second, it provides the first systematic comparison of NMR spectral features against molecular fingerprint and descriptor baselines for viscosity and surface tension prediction under identical evaluation conditions. Third, it demonstrates end-to-end fuel property prediction from mixture-level NMR spectra for a panel of conventional and alternative aviation fuels, achieving mean absolute percentage errors below 15 percent for surface tension and competitive accuracy for viscosity. Fourth, it introduces a transfer-learning framework that leverages large-scale pure-compound pretraining to improve predictions in the data-limited fuel domain. All code, data pipelines, and trained model weights are provided to support reproducibility and extension by other researchers.

2. Related Work

2.1. Classical Quantitative Structure–Property Relationships

The prediction of physical properties from molecular structure has a long intellectual history, beginning with additive and topological approaches that predate modern computing. In 1947, Harry Wiener introduced the *path number*—now called the Wiener index—defined as the sum of shortest-path distances between all pairs of non-hydrogen atoms in a molecular graph [35]. Wiener demonstrated that this single topological descriptor could predict the boiling points of paraffin hydrocarbons with surprising accuracy, establishing the foundational principle that graph-theoretic invariants encode physically meaningful structural information. Nearly three decades later, Milan Randic proposed the molecular connectivity index, computed from vertex degrees of the hydrogen-suppressed molecular graph, and showed that it correlates with boiling point, molar volume, and other properties across diverse compound classes [36]. In parallel, Corwin Hansch and Toshio Fujita formalized the linear free-energy relationship (LFER) paradigm through their ρ-σ-π analysis [37], and the subsequent Hansch–Leo compendium of substituent constants [38] provided the hydrophobic, electronic, and steric parameters that underpinned an entire generation of quantitative structure–activity and structure–property relationship models.

The group contribution (GC) approach extended these ideas to engineering property estimation. Benson and Buss formalized a hierarchy of additivity schemes—atomic, bond, and group additivity—for thermochemical properties in 1958 [39], with Benson refining these into a comprehensive tabulation in the second edition of *Thermochemical Kinetics* [40]. Although Benson group additivity was developed primarily for ideal-gas thermochemistry, it established the core principle that molecular properties can be decomposed into transferable, structure-dependent increments. Joback and Reid extended the GC paradigm to a broader set of engineering properties in 1987, proposing a unified table of 41 first-order functional groups to estimate eleven pure-component properties, including liquid dynamic viscosity [1]. Their viscosity correlation takes the Andrade form $\ln \mu = \sum_i N_i a_i - 1/T + \sum_i N_i b_i$, where N_i is the count of group i and a_i, b_i are fitted group contributions. The

Joback–Reid method became the de facto baseline for GC property estimation because of its simplicity and wide applicability, though its first-order-only decomposition cannot distinguish structural isomers. Constantinou and Gani addressed this limitation in 1994 by introducing a two-level estimation scheme, where second-order groups capture proximity effects and enable isomer differentiation [2]. Marrero and Gani further extended this architecture to three levels in 2001, adding third-order groups encoding fused ring systems and multi-functional interactions across nine properties for over 2000 compounds [41].

For mixture viscosity, the UNIFAC (UNIQUAC Functional-group Activity Coefficients) framework, originally developed by Fredenslund, Jones, and Prausnitz for vapor–liquid equilibrium [42], was adapted by Cao, Knudsen, Fredenslund, and Rasmussen to create the UNIFAC-VISCO method [43]. This approach couples the UNIFAC activity-coefficient model with an Eyring-type viscosity equation, decomposing mixture viscosity into combinatorial and residual contributions using the same VLE group-interaction parameters. The method enables, in principle, the prediction of mixture viscosity without any mixture viscosity data. Temperature dependence remains the central challenge for GC-based viscosity estimation. Most GC approaches adopt the Andrade equation, $\ln \mu = A + B/T$, where both A and B are computed as sums of group contributions. More sophisticated treatments use three-parameter forms or embed GC parameters into the Vogel–Fulcher–Tammann equation to capture the steep viscosity rise near the glass transition. Despite these refinements, GC methods for viscosity remain substantially less accurate than those for boiling points or critical constants, with typical errors ranging from 10 to 30% and degraded performance for branched, polyfunctional, or high-molecular-weight compounds where the assumption of group transferability breaks down. Modern open-source implementations, notably the Python library `chemicals` developed by Caleb Bell as part of the Chemical Engineering Design Library [32], have made Joback–Reid, Constantinou–Gani, and other GC methods programmatically accessible. The standard reference for selecting among GC methods remains *The Properties of Gases and Liquids* by Poling, Prausnitz, and O'Connell [3].

Through the 1990s and 2000s, the QSPR paradigm expanded beyond simple topological indices and group contributions to encompass large-scale molecular descriptor enumeration. Alan Katritzky and collaborators at the University of Florida developed the CODESSA (Comprehensive Descriptors for Structural and Statistical Analysis) software platform, which combined constitutional, topological, geometrical, electrostatic, quantum-chemical, and thermodynamic descriptors—over 800 in total—with automated multilinear regression and heuristic variable-selection algorithms [44]. Katritzky and Chen applied this framework directly to liquid viscosity, constructing a five-descriptor linear model for 361 diverse organic compounds that achieved $R^2 = 0.854$ [45]. A comprehensive 2010 review by Katritzky and colleagues surveyed the full landscape of structure–property correlations across dozens of physical and chemical properties [46]. In parallel, Stanton and Jurs introduced charged partial surface area (CPSA) descriptors that combined 3D molecular surface area information with electrostatic charge distributions, producing descriptors with direct physical interpretability for properties governed by intermolecular interactions such as surface tension and viscosity [47]. For surface tension specifically, Gharagheizi demonstrated that a six-descriptor linear model could predict values across 320 organic compounds with $R^2 = 0.96$ on the training set [48], while Knotts, Wilding, Oscarson, and Rowley showed that the DIPPR evaluated-data repository could serve as a high-quality training set for QSPR surface tension models [49].

The descriptor computation ecosystem matured considerably through the 2000s and 2010s. Roberto Todeschini and Viviana Consonni developed the Dragon descriptor calculator, which by version 5.4 could compute over 1,600 descriptors organized into 20 logical blocks covering constitutional, topological, geometrical, and electrostatic categories [50]. Their accompanying two-volume reference *Molecular Descriptors for Chemoinformatics* systematically catalogued over 5,000 descriptor formulations and became the de facto encyclopedia of the field [51]. Because Dragon was commercial software, open-source alternatives emerged. Chuen Wei Yap released PaDEL-Descriptor in 2011, a Java-based platform computing 797 1D/2D descriptors, 134 3D descriptors, and 10 types of fingerprints [52]. Most consequentially, Greg Landrum's RDKit—originally developed at Rational Discovery and released as open-source under a BSD license in 2006—provided a Python/C++ toolkit for descriptor calculation, substructure searching, and fingerprint generation (Morgan/ECFP, MACCS, atom pairs, topological torsions) that became the dominant platform in both academic and industrial cheminformatics [53]. The Morgan circular fingerprints [54], extended by Rogers and Hahn into the ECFP family [55], remain the most widely used molecular representations in property prediction, and their computation through RDKit has become standard practice.

Despite the successes of classical QSPR, these methods face well-documented limitations that motivate the present work. The reliance on multilinear regression assumes that property responses are linear combinations of descriptors, an assumption that fails for properties like viscosity, which depends nonlinearly on molecular size, shape, and intermolecular forces. The descriptor explosion problem—thousands of candidate descriptors versus tens or hundreds of training compounds—creates severe risks of overfitting and chance correlations, requiring careful cross-validation that was not always employed in early studies. The applicability domain of any QSPR model is inherently limited to the chemical space represented in the training data, making extrapolation to novel structural classes unreliable. Many topological descriptors are highly intercorrelated, rendering physical interpretation of multivariate models ambiguous. Perhaps most fundamentally, hand-crafted descriptors may fail to capture the structural features most relevant to a specific property. These limitations motivated the transition first to nonlinear machine-learning methods—random forests [34], gradient-boosted trees [33], and support vector machines—and ultimately to representation-learning approaches that discover task-relevant molecular features directly from structure, as discussed in subsequent sections.

2.2. Machine Learning for Molecular Property Prediction

2.2.1. Molecular Fingerprints as Feature Representations

The encoding of molecular structure as fixed-length binary or count vectors—collectively termed molecular fingerprints—provided the representational foundation on which modern machine learning for property prediction was built. The earliest widely adopted fingerprint scheme was the Molecular ACCess System (MACCS) keys, developed by MDL Information Systems in the 1980s. MACCS keys define 166 publicly available structural queries, each corresponding to a single bit that records the presence or absence of a specific substructural fragment such as an aromatic ring, a carbonyl group, or a nitrogen-containing heterocycle [56]. Because each bit position has a fixed chemical interpretation, MACCS keys are deterministic and readily interpretable, though they are limited by the finite vocabulary of fragments chosen by domain experts.

Complementary representations emerged in parallel. Carhart et al. introduced atom-pair descriptors, which enumerate every pair of atoms in a molecule along with the shortest topological path between them, capturing long-range connectivity patterns [57]. Nilakantan et al. proposed topological torsion descriptors, encoding sequences of four consecutively bonded non-hydrogen atoms together with their immediate chemical environments, thereby capturing short-range conformational information [58]. A transformative advance came with hashed path-based fingerprints, pioneered by Daylight Chemical Information Systems in the early 1990s. Rather than relying on a predefined fragment dictionary, the Daylight approach enumerates all linear bond paths up to a specified length, hashes each path to a bit position in a fixed-length vector, and folds the results together. This paradigm eliminated the need for expert curation of fragment libraries and allowed fingerprints to encode molecular features not anticipated by any predefined key set.

Building on foundations laid by Morgan's algorithm for canonical labeling of atoms at Chemical Abstracts Service [54], Rogers and Hahn formalized Extended-Connectivity Fingerprints (ECFPs), a family of circular fingerprints that iteratively aggregate atomic neighborhoods outward to a specified radius [55]. ECFP4 (radius 2, diameter 4) and ECFP6 (radius 3) became the most widely adopted molecular representations in both academic and pharmaceutical research. Unlike path-based fingerprints, circular fingerprints capture branching substructures and represent an essentially unbounded vocabulary of molecular features through hashing. In the open-source RDKit toolkit, Morgan fingerprints are generated with configurable radius (commonly 2) and bit vector length (commonly 2048 bits), and this implementation has become the de facto standard for reproducible cheminformatics research [53].

2.2.2. Fingerprints versus Molecular Descriptors

Although fingerprints were originally developed for similarity searching and virtual screening in chemical databases, their utility as feature vectors for quantitative structure–property relationship modeling became apparent as machine learning methods matured. Fingerprints offer advantages of computational speed, reproducibility, and robustness to conformational variability compared to three-dimensional descriptors [59]. Riniker and Landrum

developed an open-source benchmarking platform to systematically compare fourteen different two-dimensional fingerprints across 88 biological targets, finding that circular fingerprints—particularly Morgan and ECFP variants—consistently ranked among the top performers for ligand-based virtual screening tasks [60].

The relative merits of fingerprints and computed molecular descriptors for property prediction remain actively debated. Molecular descriptors, as catalogued comprehensively by Todeschini and Consonni [51] and implemented in packages such as Dragon [50] and PaDEL-Descriptor [52], encode continuous-valued physicochemical quantities including molecular weight, topological polar surface area, and calculated partition coefficients that fingerprints alone cannot express. Orosz et al. compared MACCS, atom-pair, and Morgan fingerprints against traditional one-dimensional, two-dimensional, and three-dimensional molecular descriptors across six ADME-Tox endpoints using XGBoost and neural network models [61]. Their findings indicated that two-dimensional molecular descriptors generally outperformed fingerprints alone, particularly with tree-based algorithms, though the margin was algorithm-dependent. In a broader review, Wigh, Goodman, and Lapkin surveyed molecular representations for machine learning and categorized fingerprints as feature-based representations that sacrifice some information content for computational efficiency and fixed dimensionality [62]. Wieder et al. further demonstrated that compact molecular fingerprints can rival more elaborate descriptor sets when paired with appropriate learning algorithms, particularly for datasets of moderate size [63]. In practice, many high-performing QSPR models concatenate fingerprints with computed descriptors, leveraging the complementary information each provides. For thermophysical property prediction specifically, fingerprints encode the presence and arrangement of functional groups that govern intermolecular interactions—hydrogen bonding, dispersion, polarity—while descriptors such as molecular weight and topological indices provide continuous-valued physicochemical context.

2.2.3. Random Forests for QSPR

The Random Forest algorithm, introduced by Breiman in 2001 [34], rapidly became one of the most widely adopted machine learning methods in QSPR modeling. Svetnik et al. were among the first to systematically evaluate Random Forests for compound classification and QSAR, demonstrating performance among the most accurate methods available at the time across six cheminformatics datasets [64]. Their work established Random Forests as the de facto baseline for molecular property prediction throughout the 2000s and early 2010s. Random Forests proved particularly well-suited to cheminformatics because they handle high-dimensional molecular descriptor spaces gracefully, provide intrinsic feature importance rankings through out-of-bag permutation measures, resist overfitting through bagging, and require minimal hyperparameter tuning. The built-in variable importance mechanism also serves as a principled feature selection tool for QSPR, enabling researchers to identify which molecular descriptors or fingerprint bits contribute most to predictive accuracy.

2.2.4. Gradient Boosting: XGBoost and LightGBM

The introduction of XGBoost by Chen and Guestrin [33] and LightGBM by Ke et al. [65] marked a substantial advance in gradient boosted decision tree performance and scalability. XGBoost introduced sparsity-aware split finding and cache-optimized computation, while LightGBM achieved further speedups through gradient-based one-side sampling and exclusive feature bundling. In molecular property prediction, Sheridan et al. provided a landmark comparison of XGBoost against Random Forests and single-task deep neural networks on proprietary QSAR datasets, finding that XGBoost with default hyperparameters consistently outperformed Random Forests and achieved accuracy nearly matching deep neural networks at dramatically faster training times [66]. This established gradient boosting as occupying a practical optimum between the simplicity of Random Forests and the complexity of deep learning, particularly for medium-sized tabular molecular descriptor datasets.

For thermophysical property prediction specifically, Chew et al. developed QSPR models for temperature-dependent viscosities of over 4000 organic molecules, finding that LightGBM with physics-informed molecular dynamics descriptors delivered the strongest descriptor-based prediction performance [67]. SHAP analysis of their models revealed that intermolecular interaction features derived from molecular dynamics simulations ranked as the most important predictors, illustrating how gradient boosted trees facilitate mechanistic interpretation through feature importance decomposition. Earlier work by Katritzky et al. had demonstrated the effectiveness of ensemble methods with CODESSA descriptors for viscosity prediction of pure organic liquids [45], and the

transition from linear and neural network QSPR to tree-based ensembles represented a consistent improvement across thermophysical property domains [46].

2.2.5. MoleculeNet Benchmarks and the Deep Learning Comparison

The MoleculeNet benchmark introduced by Wu et al. provided the first standardized comparison framework spanning quantum mechanical, physical chemistry, biophysics, and physiology tasks [68]. While learnable graph-based representations—particularly graph convolutional networks—generally offered the best performance on larger datasets, the benchmark revealed that kernel methods and ensemble tree models were more robust under data scarcity and exhibited less overfitting on small training sets. The Directed Message Passing Neural Network (D-MPNN) architecture introduced by Yang et al. through the Chemprop package advanced the state of the art further, outperforming both traditional fingerprint-based models and earlier graph neural network variants on most MoleculeNet tasks [10]. The subsequent Chemprop v2 release by Heid et al. expanded this framework with multi-task learning and uncertainty quantification capabilities [69]. However, descriptor-based models using XGBoost remained competitive in many settings, particularly when combined with well-engineered molecular descriptors. Cai et al. directly compared descriptor-based XGBoost and Random Forest models against graph neural network architectures on eleven public datasets, finding that tree-based models required only seconds to train while achieving comparable accuracy and offering superior interpretability through feature importance analysis [70].

2.2.6. When Trees Beat Deep Learning on Tabular Data

A rigorous empirical investigation by Grinsztajn et al., presented at NeurIPS 2022, demonstrated that tree-based models remain state-of-the-art on medium-sized tabular datasets—approximately 10,000 samples—even without accounting for their substantial computational advantages [71]. The study identified several properties of tabular data that favor tree-based methods: the presence of uninformative features that trees handle through implicit feature selection, irregular target functions with sharp decision boundaries that piecewise-constant tree predictions approximate well, and the lack of rotational invariance in the data that tree splits exploit naturally. These findings have direct implications for molecular property prediction with engineered descriptors. When molecular representations take the form of fixed-length descriptor vectors—whether fingerprints, RDKit descriptors, or spectral features—the prediction task is fundamentally a tabular regression problem where gradient boosted trees excel. The interpretability of these models through SHAP values and feature importance rankings provides a further practical advantage in QSPR, enabling researchers to identify which structural or spectral features drive predictions and thereby build physical intuition about structure–property relationships. For thermophysical property prediction pipelines that combine multiple descriptor types, XGBoost and LightGBM therefore represent a well-justified methodological choice that balances predictive accuracy, training efficiency, and scientific interpretability.

2.3. Deep Learning and Spectral Methods

2.3.1. Neural Networks and Learned Molecular Representations

The application of neural networks to molecular property prediction dates to the early 1990s, when multilayer perceptrons were first coupled with hand-engineered molecular descriptors for quantitative structure–property relationship modeling. Zupan and Gasteiger's 1993 textbook provided an accessible introduction that catalyzed adoption across chemical subdisciplines [72]. Around the same period, Baskin, Palyulin, and Zefirov proposed a neural architecture that bypassed explicit descriptor computation altogether, feeding graph-derived atom-level features directly into a network whose topology enforced invariance to atom renumbering [73]. These early efforts demonstrated that even shallow networks could rival or surpass linear regression when the descriptor space was sufficiently rich, but they were fundamentally limited by two constraints: the quality and completeness of the hand-crafted descriptor sets, and the modest dataset sizes typical of experimental chemistry.

A conceptual breakthrough arrived when researchers began allowing the neural network itself to learn molecular representations end-to-end, rather than relying on fixed fingerprints such as Extended-Connectivity

Fingerprints. Lusci, Pollastri, and Baldi pioneered this direction by applying recursive neural networks to undirected molecular graphs for aqueous solubility prediction, demonstrating that learned representations could outperform hand-crafted descriptors on standard benchmarks [74]. Duvenaud et al. extended this idea more generally with their "neural fingerprint" model, which replaced each layer of the circular fingerprint algorithm with a differentiable neural network layer operating on the molecular graph [75]. This work was significant not merely for its predictive accuracy but for demonstrating that circular fingerprints were a special case of a broader, learnable family of molecular representations. Concurrently, Ramsundar et al. introduced massively multitask networks for drug discovery, training a single deep network on more than 200 biological activity datasets simultaneously and showing that shared representations across tasks improved prediction on individual endpoints [76]. This was an early and influential demonstration of multi-task learning for molecular properties.

Kearnes et al. formalized the notion of molecular graph convolutions, introducing learnable edge features alongside node features and benchmarking the resulting architecture across diverse molecular property prediction tasks [77]. Their work highlighted that operating on the full molecular graph, rather than on a fixed-radius fingerprint, allowed the network to capture long-range intramolecular interactions more naturally. The conceptual landscape was then unified by Gilmer et al., who proposed the Message Passing Neural Network (MPNN) framework [9]. The MPNN framework abstracted earlier graph-based models—including Duvenaud's neural fingerprints and Kearnes's graph convolutions—into a common formalism consisting of a message function, an update function, and a readout function. Applied to the QM9 quantum chemistry benchmark, MPNNs achieved state-of-the-art results on 11 of 13 target properties. This framework became the de facto reference architecture for molecular graph neural networks, and its vocabulary—message passing, aggregation, readout—pervades the field to this day.

Sch"utt et al. developed SchNet, which introduced continuous-filter convolutional layers that operate on atomic positions in three-dimensional space rather than on a discrete bond graph [78]. By modeling interatomic distances with radial basis functions and learning filters in continuous space, SchNet could capture spatial interactions at arbitrary resolution, making it particularly effective for predicting quantum-mechanical properties of molecules and materials. Around the same time, Wu et al. released MoleculeNet, a standardized benchmark suite for molecular machine learning encompassing quantum mechanics, physical chemistry, biophysics, and physiology datasets [68]. MoleculeNet was distributed as part of the DeepChem open-source framework, which provided reference implementations of featurization pipelines and model architectures, establishing a common evaluation infrastructure that has shaped the field's trajectory.

2.3.2. Graph Neural Networks for Molecular Graphs

The year 2019 saw a rapid diversification of GNN architectures addressing distinct limitations of the original GCN and MPNN designs. Kipf and Welling's Graph Convolutional Network (GCN) had established a spectral-inspired convolutional operation on graph-structured data, where each node aggregates information from its immediate neighbors via a normalized adjacency matrix [79]. Though originally designed for citation networks, the GCN framework was quickly adopted for molecular graphs, where atoms serve as nodes and bonds as edges with atomic properties encoded as node features.

Xu et al. introduced the Graph Isomorphism Network (GIN), motivated by a theoretical analysis showing that standard GNNs are at most as powerful as the Weisfeiler–Leman graph isomorphism test in distinguishing non-isomorphic graphs [80]. GIN achieves maximal discriminative power within this bound by using a sum aggregator with learnable injective functions, making it provably as expressive as the 1-WL test. This work established a critical theoretical benchmark for GNN expressiveness that continues to shape the field, and Morris et al. extended the analysis to higher-order k-WL hierarchies [81]. In the domain of molecular fingerprints, Xiong et al. proposed AttentiveFP, which applies a graph attention mechanism to learn molecular representations directly from 2D molecular graphs [82]. AttentiveFP uses atom-level and molecule-level attention layers that weight the contributions of neighboring atoms according to learned relevance scores, enabling interpretable identification of substructures important for a given property. The model achieved state-of-the-art results across multiple MoleculeNet tasks including solubility, lipophilicity, and toxicity endpoints.

Also in 2019, Yang et al. introduced the Directed Message Passing Neural Network (D-MPNN), implemented in the Chemprop software package [10]. Unlike standard MPNNs that pass messages between atoms, D-MPNN associates messages with directed edges, which prevents so-called "totters"—redundant message paths of the form $v_1 \to v_2 \to v_1$—that dilute information flow. Benchmarked across 19 public and 16 proprietary datasets, D-MPNN consistently matched or exceeded competing architectures and random-forest baselines with molecular fingerprints, establishing Chemprop as one of the most widely adopted GNN tools in cheminformatics. Heid et al. subsequently released an updated Chemprop package extending D-MPNN with support for multi-molecule systems, reaction property prediction, uncertainty quantification, and transfer learning workflows [69].

Gasteiger et al. introduced DimeNet (Directional Message Passing for Molecular Graphs), which incorporates 3D geometric information—specifically interatomic distances and bond angles—into the message passing framework [83]. By embedding messages between atom pairs and transforming them based on angular information, DimeNet captures directional interactions that are invisible to purely topological 2D GNNs, achieving substantial improvements on molecular energy and force prediction tasks. DimeNet and its faster successor DimeNet++ demonstrated that encoding 3D structure is essential for quantum-chemical property prediction, though at the cost of requiring 3D conformer generation as a preprocessing step. Hu et al. extended the benchmarking effort with the Open Graph Benchmark, which includes molecular datasets derived from MoleculeNet but with more rigorous scaffold-based splits and a leaderboard infrastructure that has driven rapid progress [84].

Despite their success, GNNs face several well-characterized limitations. The over-smoothing problem causes node representations to converge to indistinguishable states as the number of message-passing layers increases, effectively limiting GNN depth and the size of molecular neighborhoods that can be captured. The related phenomenon of over-squashing compresses information from exponentially growing receptive fields into fixed-size vectors, losing long-range dependencies. Wieder et al. catalogued over 80 GNN variants applied to more than 20 molecular properties across 48 datasets, confirming the breadth of GNN adoption while highlighting these persistent challenges [63]. For thermophysical property prediction specifically, the requirement for large, high-quality training data remains a bottleneck: viscosity and surface tension datasets are orders of magnitude smaller than the drug-discovery datasets on which most GNN architectures were developed. Additionally, temperature and pressure dependence—critical for thermophysical applications—requires either multi-task formulations or auxiliary input features, as standard molecular GNNs encode only molecular structure and not thermodynamic state variables.

2.3.3. Transformers and Self-Supervised Pretraining

The success of the Transformer architecture in natural language processing prompted researchers to investigate whether molecular strings could be treated as a chemical "language." Schwaller et al. provided an early and influential demonstration with the Molecular Transformer, which cast chemical reaction prediction as a sequence-to-sequence translation problem over SMILES strings, achieving over 90% top-1 accuracy on standard benchmarks without requiring hand-crafted atom mappings [85]. Honda et al. introduced the SMILES Transformer, which used unsupervised encoder–decoder pretraining on a large SMILES corpus to learn continuous molecular fingerprints [86]. This work showed that pretrained representations outperformed rule-based fingerprints and graph methods specifically in low-data regimes, a finding directly relevant to property prediction tasks with limited labeled compounds.

The bidirectional masked language modeling paradigm pioneered by BERT was quickly adapted for chemistry. Fabian et al. proposed MolBERT, which combined masked SMILES token prediction with domain-relevant auxiliary tasks such as molecular descriptor regression, demonstrating that injecting chemical knowledge during pretraining improved downstream virtual screening and QSAR performance even with a relatively modest four-million-compound pretraining set [87]. Chithrananda et al. released ChemBERTa, adapting the RoBERTa architecture to 77 million PubChem SMILES with purely self-supervised masked language modeling, showing that performance scaled predictably with pretraining corpus size [88]. Ahmad et al. extended this line with ChemBERTa-2, systematically comparing masked language modeling against multi-task supervised pretraining and establishing that multi-task pretraining on computed descriptors yielded the most transferable representations [89]. Beyond token-level masking, contrastive self-supervised methods emerged as a complementary strategy. Wang et al. introduced MolCLR, which operates on molecular graphs rather than SMILES strings, applying

stochastic augmentations—atom masking, bond deletion, subgraph removal—and training a GNN encoder with a contrastive objective that pulls augmented views of the same molecule together while pushing different molecules apart [90].

A parallel thread addressed a fundamental limitation of SMILES-based models: the SMILES string is a linearization of molecular topology and discards explicit 3D spatial information. Ying et al. introduced Graphormer, a Transformer architecture for graph-structured data that encodes structural information through centrality encoding, spatial encoding based on shortest-path distances, and edge feature encoding in the attention mechanism [91]. Graphormer won first place in the OGB-LSC quantum property prediction challenge and demonstrated that standard Transformer attention, when properly augmented with graph-structural biases, can outperform specialized message-passing GNNs. Rampasek et al. generalized this direction with the GPS (General, Powerful, Scalable) framework, which decouples local message-passing from global Transformer attention, achieving linear complexity in the number of nodes and edges while matching or exceeding Graphormer's performance across 16 benchmarks [92]. Moving further toward physical realism, Zhou et al. proposed Uni-Mol, a universal 3D molecular pretraining framework that directly takes atomic 3D coordinates as input using an SE(3)-equivariant Transformer backbone [93]. Pretrained on 209 million molecular conformations with masked atom prediction and 3D position denoising tasks, Uni-Mol outperformed prior state-of-the-art on 14 of 15 molecular property prediction benchmarks, confirming that 3D geometric information provides substantial gains over 1D or 2D representations alone.

2.3.4. Transfer Learning for Small Datasets

A persistent challenge in molecular property prediction is label scarcity: experimental measurements are expensive, and many target properties have only hundreds or low thousands of data points. Hu et al. systematically addressed this by evaluating strategies for pretraining graph neural networks [94]. They showed that naive node-level or graph-level pretraining could lead to negative transfer, but a combined strategy—pretraining at both levels using self-supervised objectives on large unlabeled molecular corpora, followed by fine-tuning on the target task—yielded up to 9.4% absolute improvement in ROC-AUC on molecular property benchmarks. This work established pretraining as a viable strategy for molecular GNNs and stimulated a wave of follow-up research on self-supervised molecular representation learning.

These developments have collectively pushed the field toward chemical foundation models: large pretrained architectures intended to serve as general-purpose molecular encoders [95]. The central value proposition is that self-supervised pretraining on millions of unlabeled structures learns transferable chemical representations that can be fine-tuned with far fewer labeled examples, directly addressing the data scarcity that characterizes most experimental property prediction tasks. For a problem such as the one considered in this report, where fewer than 1,000 compounds have matched NMR spectra and thermophysical measurements, the pretrain-then-fine-tune paradigm is especially compelling: the model can learn general molecular syntax, substructure patterns, and conformational preferences from vast unlabeled corpora, then adapt to the specific NMR-to-property mapping with the limited labeled data available.

However, significant limitations remain. SMILES-based models are sensitive to tokenization choices and SMILES canonicalization. Graph Transformers require careful positional encoding design to capture the structural information that message-passing GNNs handle naturally. 3D pretraining methods such as Uni-Mol depend on the quality of conformer generation, which introduces noise for flexible molecules. Recent analyses have noted that foundation models for chemistry still struggle with out-of-distribution generalization and that the benefits of scale plateau when downstream tasks involve property types—such as bulk thermophysical properties—that are poorly represented in pretraining objectives focused on quantum-level or pharmacological endpoints [95]. For thermophysical applications specifically, GNN and Transformer models have been applied to viscosity prediction by Chew et al. [67] and to surface tension of polymers by Tamnanloo et al. [96], but both studies found that incorporating physics-based features or domain-specific descriptors substantially improved predictions beyond what end-to-end learned representations achieved alone. These considerations motivate hybrid approaches that combine experimentally grounded spectroscopic descriptors with learned representations rather than relying solely on pretrained models.

2.4. NMR and Machine Learning

2.4.1. First-Principles NMR Prediction

The computational prediction of NMR chemical shifts from molecular structure has a long history in quantum chemistry and provides the foundation against which all subsequent machine-learning approximations are benchmarked. Wolinski, Hinton, and Pulay introduced the gauge-including atomic orbital (GIAO) method in 1990, enabling density functional theory (DFT) level calculation of NMR shielding tensors for isolated molecules by resolving the gauge-origin problem that arises in finite basis set calculations [97]. For periodic solids, Pickard and Mauri extended first-principles NMR prediction to crystalline systems through the gauge-including projector augmented wave (GIPAW) method, which operates under periodic boundary conditions and has become the standard approach for solid-state NMR calculations [98]. Together, GIAO and GIPAW form the two pillars of ab initio NMR prediction. However, both approaches carry substantial computational cost. GIAO calculations scale steeply with basis set size and molecular complexity, often requiring hours per molecule even on modern hardware. GIPAW calculations on molecular crystals with hundreds of atoms in the unit cell can demand days of wall time on computing clusters. This prohibitive expense motivated the search for faster alternatives that could retain chemical accuracy while enabling high-throughput screening of thousands or millions of compounds.

2.4.2. Machine Learning for Chemical Shift Prediction

The first generation of machine-learning models for NMR chemical shift prediction sought to reproduce DFT-quality results at a fraction of the computational cost. Jonas and Kuhn developed a deep neural network with quantified uncertainty estimates that achieved root-mean-square errors of 1.43 ppm for ^{13}C and 0.28 ppm for ^{1}H, both within or near DFT accuracy, while running orders of magnitude faster than quantum chemical methods [99]. Their model's built-in uncertainty quantification represented an important advance, as it allowed users to assess the reliability of individual predictions rather than relying solely on aggregate error statistics.

Paruzzo et al. introduced ShiftML, a Gaussian process regression model trained on GIPAW-computed shifts for approximately 2,000 molecular solids selected by farthest-point sampling from a database of over 60,000 crystal structures [100]. By encoding local chemical environments with smooth overlap of atomic positions (SOAP) descriptors, ShiftML achieved R^2 values of 0.97 for ^{1}H, 0.99 for ^{13}C, 0.99 for ^{15}N, and 0.99 for ^{17}O chemical shifts, all within DFT accuracy, while accelerating predictions by up to six orders of magnitude for the largest structures. The success of ShiftML demonstrated that local atomic environment descriptors, originally developed for interatomic potentials, could be repurposed for spectroscopic property prediction with remarkable fidelity.

Graph-based architectures subsequently emerged as a dominant paradigm for solution-state NMR prediction. Kwon et al. applied message passing neural networks (MPNNs) operating on molecular graphs with implicit hydrogen representation, demonstrating high accuracy for both ^{1}H and ^{13}C shifts [101]. Yang, Chakraborty, and White extended this line of work with a graph neural network framework that captured hydrogen-bonding-induced downfield shifts and protein secondary structure effects, establishing GNNs as a versatile tool for NMR prediction across small molecules and biomolecules alike [102]. These graph-based models offer a conceptually natural representation for molecules: atoms become nodes, bonds become edges, and message passing aggregates local chemical environment information in a manner analogous to the physical interactions that determine chemical shifts.

2.4.3. Structure Elucidation and Spectral Interpretation

While the models described above solve the *forward* problem of predicting NMR observables from known structures, a parallel and arguably more challenging line of research tackles the *inverse* problem: deducing molecular structure from NMR spectra. Computer-assisted structure elucidation (CASE) systems have been under development for over five decades, but their integration with modern machine learning has accelerated markedly in recent years. Elyashberg reviewed the state of the field in 2021, noting that contemporary CASE systems exploit one- dimensional and two-dimensional NMR data in combination with DFT-computed reference shifts and are capable of elucidating complex natural products, though they still rely heavily on expert-curated rules and spectral databases [103].

More recently, deep learning has begun to replace or augment these rule-based pipelines. Wei et al. developed pseudo-Siamese convolutional neural networks (pSCNN) that achieved 99.8% accuracy for compound identification from one-dimensional NMR spectra of mixtures, demonstrating that raw spectral vectors could serve as effective inputs for convolutional architectures [104]. Li et al. introduced 2DNMRGym, an annotated experimental dataset for atom-level molecular representation learning from two-dimensional NMR spectra via surrogate supervision, providing a benchmark for training models that bridge spectral data and molecular structure [105]. These efforts reflect a broader trend toward end-to-end learning from spectral data, moving away from the traditional pipeline of peak picking, assignment, and rule-based interpretation toward architectures that learn directly from the spectral signal.

2.4.4. NMR Metabolomics and Automated Profiling

In metabolomics, automated compound identification and quantification from complex NMR mixture spectra has long been a bottleneck limiting throughput. Ravanbakhsh et al. introduced BAYESIL, the first fully automated system for quantitative ^{1}H NMR spectral profiling, which framed metabolite identification as probabilistic inference within a graphical model and achieved approximately 90% correct identification with approximately 10% quantification error in under five minutes [106]. The Human Metabolome Database (HMDB), which has grown to contain over 220,000 metabolite entries with associated NMR reference spectra, provides the foundational spectral library against which such automated systems operate [11].

Subsequent deep learning approaches improved upon the BAYESIL foundation. The pseudo-Siamese CNN architecture of Wei et al. demonstrated that mixture spectra could be deconvolved and individual compounds identified with near-perfect accuracy [104], while one-dimensional CNN architectures applied to low-field NMR transverse relaxation signals achieved robust classification of edible oils and other complex mixtures. These metabolomics applications established a critical precedent for the present work: they proved that minimally processed NMR spectral vectors, treated as fixed-length numerical feature vectors, could serve as effective inputs for supervised learning models. The spectral featurization strategies developed in the metabolomics context—including bucket integration, peak alignment, and normalization—translate directly to the problem of encoding NMR spectra as molecular descriptors for property prediction.

2.4.5. The Gap: NMR Spectra as Features for Property Prediction

Despite the maturity of machine learning methods for NMR shift prediction (structure to spectrum) and structure elucidation (spectrum to structure), remarkably few studies have explored using NMR spectral representations to predict bulk physical or thermophysical properties such as viscosity, density, surface tension, or heat capacity. The overwhelming majority of molecular property prediction models rely on structural fingerprints (ECFP, MACCS), computed descriptors (RDKit, Dragon), SMILES strings, or molecular graphs as input representations. NMR spectra, whether experimental or predicted, have been almost entirely absent from this literature.

Leniak, Pietrus, and Kurczab published one of the first explicit investigations of this direction in 2024, designing NMR-derived chemical representations for predicting logD using support vector regression and gradient boosting methods [107]. Their approach discretized predicted ^{1}H NMR spectra into bucket-integrated feature vectors and achieved RMSE of 0.66, competitive with models trained on conventional molecular fingerprints. In a follow-up study, the same group fused predicted ^{1}H and ^{13}C spectra into compact descriptor vectors and trained one-dimensional CNNs that matched the performance of traditional ECFP4 fingerprints at one-fifth the dimensionality [108]. Their work demonstrated that NMR-derived representations carry sufficient structural information to support quantitative structure-property relationship modeling, though they focused exclusively on pharmaceutical properties (logD, solubility) rather than thermophysical quantities.

In parallel, Zipoli, Alberts, and Laino released a large-scale computational IR–NMR multimodal spectral dataset covering 177,000 patent-extracted organic molecules [109], providing the kind of large, standardized spectral dataset that data-driven property prediction models require. The availability of such datasets, combined with the growing repositories of experimental NMR data from HMDB [11] and NMRShiftDB [110], creates the infrastructure necessary for training models that map spectral features to physical properties at scale.

The gap that remains is clear. Abundant work exists on predicting NMR spectra from structure, and substantial progress has been made on deducing structure from NMR spectra. However, the direct path from NMR spectral features to thermophysical property values—bypassing explicit structural identification—remains largely unexplored. This gap is significant because NMR spectra encode rich information about molecular topology, functional group environments, and conformational preferences that conventional molecular fingerprints may fail to capture. A ^{1}H NMR spectrum, for instance, simultaneously encodes the presence and relative abundance of aromatic, olefinic, aliphatic, and heteroatom-adjacent proton environments in a single measurement. These are precisely the structural features that govern intermolecular interactions and, consequently, bulk transport and interfacial properties. The present work addresses this gap by treating NMR spectral vectors—both experimental and predicted—as molecular descriptors for viscosity, density, and surface tension regression, bridging the mature NMR-machine-learning literature with the thermophysical property prediction domain discussed in the preceding sections.

2.5. Fuel Property Prediction from Spectroscopy

The application of spectroscopic methods to fuel property prediction has roots in the chemometrics revolution of the 1980s, when partial least squares (PLS) regression and principal component regression (PCR) were first coupled with infrared and NMR spectra to estimate hydrocarbon fuel properties. Among the earliest demonstrations was the work of Gulder [111], who showed that proton NMR spectroscopy could predict the cetane number of diesel fuels from carbon-type structural composition determined by ^{1}H spectral integration. By the mid-1990s, Fodor et al. [112] had established a comprehensive protocol for the simultaneous determination of gasoline properties—including research and motor octane numbers, aromatic and olefinic content, and oxygenate concentrations—from a single FT-IR spectrum using multivariate calibration, achieving analysis times under one minute with less than two milliliters of sample. These foundational studies demonstrated that vibrational and magnetic resonance spectra encode sufficient molecular information to replace time-consuming wet-chemical methods, and PLS remains one of the most widely used regression techniques for fuel property prediction. The ASTM standards that define fuel quality—D1655 for aviation turbine fuel [23], D7566 for synthetic blending components in jet fuel [24], and D341 for viscosity–temperature relationships [26]-–require extensive physical testing across dozens of property specifications, creating strong incentive for rapid spectroscopic alternatives.

Near-infrared (NIR) and mid-infrared (MIR) spectroscopy have become the workhorses of online fuel quality monitoring in refineries. NIR instruments offer nondestructive, rapid analysis of parameters including density, viscosity, distillation curve, sulfur content, cold filter plugging point, and oxidation stability, making them well suited for both at-line quality control and in-line blending optimization. Baird and Oja [113] reviewed the chemometrics literature and extended NIR-based prediction to temperature-dependent density, achieving relative errors below one percent using support vector regression. Mid-infrared spectroscopy provides complementary information through functional-group-specific absorption features, and has been applied to diesel quality indicators with PLS models. Raman spectroscopy has emerged as a third vibrational technique for fuel characterization, offering advantages of minimal sample preparation, compatibility with aqueous environments, and analysis times as short as five seconds. Raman has proven effective for gasoline octane rating, boiling point prediction, and detection of adulterated fuels, and is now considered an emerging standard in oil, gas, and petrochemical analytical laboratories. Collectively, the vibrational spectroscopy literature establishes that multivariate regression on spectral data can approximate standard ASTM test methods, but these techniques primarily probe bulk vibrational modes and struggle to distinguish structural isomers or resolve individual carbon environments within complex hydrocarbon mixtures.

Nuclear magnetic resonance spectroscopy provides a uniquely atom-resolved description of molecular structure that complements the bulk functional-group information available from IR and Raman. The Clean Combustion Research Center at KAUST, led by Sarathy and Abdul Jameel, has pioneered the use of ^{1}H NMR for fuel property prediction through a systematic program of publications spanning 2016–2021. Abdul Jameel et al. [18] developed an improved model for derived cetane number (DCN) prediction using ^{1}H NMR structural parameters—paraffinic CH_3, CH_2, and CH groups, naphthenic CH–CH_2, olefinic CH=CH_2, and aromatic C–CH moieties—achieving R^2 values of 0.97 via multiple linear regression across a diverse set of diesel fuels. Their subsequent minimalist functional group (MFG) framework [20] generalized this approach to surrogate fuel formulation for both diesel and

jet fuels, defining a compact set of NMR-derived functional group concentrations that could reproduce ignition quality tester measurements and smoke point data. Abdul Jameel et al. [19] extended the approach to octane number prediction using artificial neural networks trained on NMR spectra, demonstrating that nonlinear models could capture interactions among functional groups that linear methods miss. A 2021 study [114] further refined the identification and quantification of hydrocarbon functional groups in gasoline, expanding the palette of NMR-derived descriptors available for property prediction. Kapur et al. [115] had earlier demonstrated, independently, that neural networks could improve upon linear NMR–cetane models by capturing nonlinear structure–property relationships, anticipating the machine learning approaches that would later become standard. Cookson and Smith [16] provided still earlier groundwork by showing that ^{13}C NMR could resolve the saturate fraction of diesel and kerosene fuels into detailed structural categories, establishing the spectral interpretation framework on which later quantitative models were built.

The convergence of modern machine learning with spectroscopic data has accelerated fuel property prediction beyond the capabilities of classical chemometrics. Vozka and Kilaz [116] reviewed aviation turbine fuel composition–property relations across multiple analytical platforms—NMR, IR, Raman, and gas chromatography—and concluded that spectroscopic methods consistently outperform empirical correlations when the training data span sufficient compositional diversity. Moro et al. [117] cataloged chemometric models for predicting crude oil properties from both NMR and IR spectroscopy, confirming the broad applicability of multivariate spectral regression across the petroleum value chain while noting that most published models rely on small datasets of 20–100 samples and may not generalize to novel fuel compositions. More recent work has pushed beyond PLS toward gradient-boosted trees and neural networks. Parker et al. [21] demonstrated that NMR atom-type descriptors combined with XGBoost can predict liquid density and surface tension of pure hydrocarbons with accuracy rivaling group contribution methods, and that composition-weighted NMR descriptors transfer effectively to real fuel mixtures. Zipoli et al. [109] compiled a multimodal IR–NMR dataset of 177,000 patent-extracted organic molecules, providing the scale of training data needed for deep learning approaches to spectral property prediction.

These methodological advances address a critical bottleneck in the deployment of sustainable aviation fuels. The ASTM D4054 qualification pathway for new SAF production routes has historically required four tiers of testing spanning five or more years and costing millions of dollars [29]. Although the Fast Track provision permits abbreviated testing for fuels compositionally similar to conventional Jet-A, blend limits are capped at ten percent, and only about eleven pathways have achieved full D7566 approval as of 2024 [28]. Heyne et al. [118] outlined prescreening tools and procedures that use low fuel volumes to estimate critical properties early in the development process, arguing that rapid, low-cost analytical methods could dramatically reduce the time and cost of SAF qualification. The Air Force Research Laboratory maintains the most comprehensive database of jet fuel properties, documented in Edwards [22] and the subsequent 2020 technical report [119], covering density, viscosity, distillation, derived cetane number, surface tension, freeze point, flash point, and heat of combustion for hundreds of conventional and alternative fuels identified by POSF numbers. This database, together with complementary resources from the National Jet Fuels Combustion Program [25], provides the ground-truth measurements against which spectroscopic models must be validated.

NMR-based approaches are particularly promising for SAF screening because they provide direct, quantitative structural information—functional group distributions, branching indices, and aromatic character—that correlates physically with macroscopic thermophysical properties. Unlike IR and Raman, which primarily probe vibrational modes, NMR resolves individual carbon and hydrogen environments and can distinguish structural isomers that exhibit very different fuel performance characteristics. A branched C_{12} isoparaffin and its linear isomer, for example, have nearly identical IR spectra but markedly different viscosities and freeze points; their ^{1}H NMR spectra, by contrast, clearly resolve the methyl branching pattern. Current limitations include the capital cost of high-field NMR instrumentation and the need for larger, more diverse training sets that span the compositional space of novel SAF candidates. The development of benchtop NMR spectrometers operating at 40–80 MHz is rapidly reducing the cost barrier, bringing per-sample analysis costs to levels competitive with gas chromatography. When combined with the growing availability of large-scale spectral databases and modern machine learning architectures, NMR-based property prediction offers a path toward the rapid, low-volume fuel screening tools that the SAF qualification process urgently requires.

3. Data and Datasets

3.1. NMR Spectral Databases

The foundation of any data-driven approach to NMR-based property prediction is a large, structurally diverse corpus of molecular NMR spectra paired with unambiguous chemical identifiers. This section surveys the fourteen NMR spectral databases evaluated for this work, ranging from million-scale text-mined collections to small, expertly curated reference libraries. Table 1 provides a summary of all databases considered, including their scale, data format, and access method. The databases span a wide range of provenance: some contain exclusively experimental measurements, others provide computationally predicted spectra, and several offer both. They also differ dramatically in data representation, from raw free induction decays (FIDs) to processed peak lists to bucketed pseudo-spectra, and in accessibility, from fully open REST APIs to Cloudflare-protected web interfaces with no programmatic access whatsoever.

Database	**Compounds**	**Spectra**	**Format**	**Access Method**
NMRexp	1,480,000	3,373,000	Parquet peak lists	Zenodo download
NMRShiftDB2	45,000	70,000	SDF / NMReDATA	SourceForge bulk export
SDBS	34,600	30,100[a]	HTML images + metadata	Web scraping
NP-MRD	281,859	5,400,000[b]	nmrML / JCAMP-DX / FID	Bulk FTP download
HMDB	217,920	313,380[c]	XML / FID / peak lists	Bulk download[d]
BMRB/GISSMO	3,315	90,000[e]	STAR / peak tables	REST API / FTP
nmrXiv	—[f]	151 projects	Raw FID (vendor)	REST API
NMRBank	149,135	225,809	CSV peak lists	GitHub download
2DNMRGym	22,348	22,348	HSQC peak tables	GitHub / HuggingFace
PROSPRE	577	4,207 shifts	Web / downloadable	Web server
IR-NMR	1,255[g]	1,255	Parquet (DFT shifts)	Zenodo download
logD-NMR	1,276	1,276	CSV (200-bin vectors)	GitHub download
BML-NMR	208	3,328	FID / processed	Offline[h]
PubChem	116,000,000	—	SMILES / InChI	FTP bulk download

Table 1: Summary of NMR spectral databases surveyed in this work. [a]Combined ^{1}H and ^{13}C NMR entries. [b]Includes 5M+ predicted spectra. [c]Approximately 400 experimental, remainder predicted. [d]Cloudflare WAF requires workaround. [e]Peak annotations from GISSMO spin simulations across 18 field strengths. [f]Compound count not directly reported; data organized by project. [g]NMR subset of 177,461-compound IR dataset. [h]Server unreachable since September 2025.

3.1.1. NMRexp

NMRexp is the largest publicly available database of experimental NMR spectra for organic molecules, published by Wang et al. in *Scientific Data* in December 2025 [13]. Developed at Peking University and the AI for Science Institute in Beijing, NMRexp contains 3,372,987 experimental NMR records spanning six nuclei: ^{1}H (1,667,135 spectra), ^{13}C (1,455,670), ^{19}F (197,186), ^{31}P (33,456), ^{11}B (17,143), and ^{29}Si (2,397). Among the ^{1}H spectra alone, 1,417,781 unique compounds are represented, with approximately 8% of compounds having two or more independent measurements. The dataset was constructed by text-mining nearly 200,000 supporting information PDF documents from scientific publications spanning 2010 to 2024. The extraction pipeline employs a YOLO-based layout detection model to identify molecular structure diagrams and text blocks, an optical chemical

structure recognition module to convert molecular images to SMILES strings, and approximately 60 regular expressions to parse NMR peak data. Each entry undergoes validation against fundamental NMR principles, including proton count consistency, monotonic chemical shift ordering, and plausible coupling constant ranges.

The dataset is distributed as Parquet files on Zenodo under a CC-BY-4.0 license. The primary file (661 MB) contains columns for SMILES, NMR type, spectrometer frequency, solvent, normalized peak text, structured peak lists with chemical shifts and multiplicities, and source DOIs for traceability. A filtered subset (529 MB, 2,853,590 entries) restricts to molecules with at most one stereogenic carbon. Manual validation of 300 randomly sampled entries confirmed greater than 99% accuracy in metadata extraction and 98% correctness in molecular skeleton assignment. Internal consistency tests on replicate spectra yielded mean absolute errors of 0.026 ppm for ^{1}H and 0.206 ppm for ^{13}C, substantially lower than the 0.09 ppm and 0.51 ppm reported for NMRShiftDB2. Known limitations include optical structure recognition errors in approximately 2% of molecular skeletons and stereochemistry recognition accuracy of only 83% for molecules with a single stereocenter. NMRexp surpasses existing open-access NMR databases by one to two orders of magnitude in scale, making it comparable to commercial databases while remaining fully open.

3.1.2. NMRShiftDB2

NMRShiftDB2 is the longest-running open-access database of experimentally measured NMR chemical shifts, tracing its origins to the NMRShiftDB project initiated by Steinbeck and Kuhn at the Max-Planck-Institut in Jena in 2002 [120], [121]. The original system went live in November 2003, and by mid-2004 it held over 10,000 spectra. Primary hosting now resides at the NMR laboratory at the University of Cologne under Nils Schloerer. As of early 2026, the database reports 271,816 structures, 70,029 measured spectra, and 396,583 calculated spectra. In terms of unique experimentally measured entries, the database encompasses approximately 54,000 spectra across roughly 45,000 distinct molecules, covering ^{1}H, ^{13}C, ^{19}F, and other nuclei. A distinguishing feature is its peer-review mechanism: all submitted datasets are reviewed by a scientific advisory board before inclusion, lending the data a level of reliability uncommon among open chemical databases.

Bulk data exports are distributed via SourceForge in several formats. The NMReDATA SDF file (284 MB) provides the richest representation, encoding assigned spectra in the NMReDATA standard. A signals-annotated SDF (159 MB) and a basic structures-only SDF (139 MB) are also available, along with 3D-coordinate variants and compressed PostgreSQL dumps (1.1 GB). The database is integrated into the NFDI4Chem initiative in Germany, where it serves as a curated NMR data repository component. NMRShiftDB2 has been used extensively as a benchmark dataset for NMR chemical shift prediction algorithms, including hierarchical substructure-based (HOSE code) predictors, graph neural networks, and deep learning approaches, making it the de facto standard reference for evaluating shift prediction accuracy.

3.1.3. SDBS

The Spectral Database for Organic Compounds (SDBS) is one of the oldest and most respected freely available spectral reference databases in analytical chemistry, maintained by the National Metrology Institute of Japan (NMIJ) within AIST [14], [15]. Research activities underlying SDBS began in 1982, with systematic in-house measurement of authentic spectral data to ensure analytical reliability. Digital access was established in 1989, followed by free public web-based release in 1997. As of March 2025, SDBS catalogs approximately 34,600 organic compounds with six types of spectra: ^{1}H NMR (approximately 15,900 spectra), ^{13}C NMR (approximately 14,200 spectra), Fourier-transform infrared (approximately 54,100 spectra), electron ionization mass spectra (approximately 25,000), laser Raman (approximately 3,500), and electron paramagnetic resonance (approximately 2,000). Two-thirds of the compounds have between 6 and 16 carbon atoms, reflecting the focus on commercial organic reagents. All spectra were measured in-house by AIST researchers using calibrated instruments, lending the data exceptionally high reliability.

Despite its scientific value, SDBS presents significant challenges for programmatic data access. The database offers no bulk download mechanism, no public API, and operates exclusively through a web interface that enforces a click-through disclaimer agreement. Session-based access controls and rate limiting further impede automated retrieval. SDBS does not provide SMILES strings for its compounds; instead, it offers compound names, CAS

registry numbers, molecular formulae, and InChI identifiers. Resolving these to canonical SMILES requires secondary lookups against external services such as PubChem. For this project, custom scraping infrastructure was developed using Playwright browser automation and direct HTTP requests with parallelized workers to enumerate compound metadata. A fallback lookup to PubChem by CAS number was used to obtain canonical SMILES for compounds lacking InChI in SDBS.

3.1.4. NP-MRD

The Natural Products Magnetic Resonance Database (NP-MRD) is the world's largest freely available NMR database dedicated to natural products, maintained by Wishart and collaborators at the University of Alberta with support from NIH [122]. As of its current release, NP-MRD contains 281,859 natural product compounds encompassing small molecules derived from plants, fungi, bacteria, marine organisms, and animals. The database stores over 5.4 million total NMR spectra, of which approximately 19,045 are experimental (18,259 1D and 786 2D), 416,405 are simulated from experimental assignments, and over 5 million are computationally predicted using the CASPRE/PROSPRE prediction engines. Data are distributed in multiple formats including nmrML, JCAMP-DX, raw FID files in Bruker TopSpin format, chemical shift peak lists, and compound metadata in XML and JSON. The full download archive spans well over 100 GB, with the nmrML spectral files alone exceeding 160 GB. NP-MRD integrates data from several partner databases, including HMDB, BMRB, the JEOL CH-NMR-NP collection, and the Natural Products Atlas, with all data released under a CC BY-NC 4.0 license.

NMR data for natural products differ from those of synthetic compounds in several important respects. Natural products typically exhibit far greater structural diversity and stereochemical complexity than synthetic molecules, with extensive ring systems, chiral centers, and unusual functional groups (terpenoids, polyketides, alkaloids, glycosylated metabolites) that produce highly congested and overlapping spectra. Solvent variability is a major practical concern, as natural products are often characterized in $CDCl_3$, DMSO-d_6, CD_3OD, D_2O, or pyridine-d_5 depending on solubility. NP-MRD addresses this by storing rich metadata including species of origin, chemical ontology classifications, and literature references for every compound, and by providing DFT-calculated chemical shift predictions within 24 hours of deposition to aid in structure validation. The database also provides ML-predicted ^{1}H and ^{13}C spectra at multiple spectrometer frequencies and includes integrated spectral comparison utilities.

3.1.5. HMDB

The Human Metabolome Database (HMDB) is the most comprehensive freely available electronic database dedicated to the human metabolome, developed and maintained by the Wishart Research Group at the University of Alberta [11], [123]. Since its initial release as HMDB 1.0 in 2007 with 2,180 metabolite entries, the database has undergone exponential growth through four major revisions: HMDB 2.0 (2009, 6,800 entries), HMDB 3.0 (2013, 40,153 entries) [124], HMDB 4.0 (2018, 114,100 entries) [125], and the current HMDB 5.0 (2022, 217,920 entries). Beyond metabolite enumeration, HMDB provides extensive chemical, clinical, and molecular biological annotations including chemical taxonomy, physiological concentrations, disease associations, and pathway information.

Of particular relevance to this work, HMDB contains both experimental and computationally predicted NMR spectral data. The experimental NMR collection includes approximately 400 metabolite standards with high-quality ^{1}H and ^{13}C NMR spectra acquired under controlled conditions, along with associated raw FID files. HMDB 5.0 substantially expanded its predicted spectral coverage, adding 312,980 predicted ^{1}H and ^{13}C NMR spectra computed using density functional theory and machine learning methods. Accessing HMDB data programmatically presents practical challenges: the server employs Cloudflare Web Application Firewall protection that blocks automated HTTP requests, returning 403 Forbidden responses even with browser-mimicking user agents. For this project, the primary metabolite data file (910 MB XML dump with SMILES for all 217,920 entries) was obtained via the Internet Archive's Wayback Machine, along with HMDB 2.5-era NMR archives containing FIDs, peak lists, and processed spectra totaling 2.3 GB.

3.1.6. BMRB and GISSMO

The Biological Magnetic Resonance Data Bank (BMRB) was founded in 1988 by Markley and Ulrich at the University of Wisconsin–Madison as the international open-data repository for biomolecular NMR spectroscopy [126]. Originally focused on protein and nucleic-acid chemical shifts, BMRB became a core archive of the Worldwide Protein Data Bank in 2007 and now houses over 11 million assigned chemical shifts from more than 15,000 deposited studies. Operations transferred to UConn Health in 2020, though the archive retains deep ties to the National Magnetic Resonance Facility at Madison (NMRFAM). A dedicated metabolomics wing provides experimental time-domain data, reference spectra, and standardized chemical identifiers for more than 1,200 common small-molecule metabolites, making it the primary public resource for NMR-based metabolite identification in biofluids, natural-product screening, and clinical biomarker discovery.

The GISSMO (Guided Ideographic Spin System Model Optimization) library, developed as a collaboration between NMRFAM, NMRBox, and BMRB, extends the metabolomics archive from static peak lists to full quantum-mechanical spin-system parameterizations [127]. For each compound, GISSMO refines a matrix of chemical shifts and scalar coupling constants against an experimental 1D ^{1}H spectrum, then uses that matrix to simulate spectra at 18 magnetic-field strengths spanning 40 to 1,300 MHz. The current library contains more than 1,286 parameterized compounds, including 656 endogenous metabolites, 665 drug-like fragments from the Maybridge Ro3 screening library, and 487 lignin derivatives, with peak lists exported for every simulated spectrum. In the dataset assembled for this work, GISSMO entries were parsed into a relational table keyed on BMRB entry identifier and linked to SMILES strings, yielding approximately 3,315 unique structures and roughly 90,000 individual ^{1}H peak annotations. The spin-system matrices provide a field-strength-independent representation that generalizes across spectrometer configurations.

3.1.7. nmrXiv

nmrXiv is an open, FAIR-compliant repository for NMR data developed under the NFDI4Chem initiative in Germany [128]. Unlike legacy NMR databases such as SDBS or HMDB, which store only processed peak lists or static spectral images, nmrXiv is designed to host raw, instrument-level NMR data, including original FIDs in vendor-native formats (Bruker, JEOL, Varian) alongside processed spectra in open formats such as JCAMP-DX and NMReDATA. Each deposition is assigned a persistent DOI, machine-readable metadata conforming to Schema.org conventions, and a Creative Commons license, ensuring long-term discoverability and reuse. The platform enforces FAIR principles at the repository level by requiring structured metadata, open-access APIs, and standardized chemical ontologies for compound annotation. As of early 2026, nmrXiv hosts hundreds of curated projects spanning organic, natural-product, and synthetic chemistry.

For the present work, 151 publicly available nmrXiv projects were downloaded programmatically through the platform's REST API, which exposes project, study, and dataset endpoints in JSON format. Each project contains one or more studies, and each study contains one or more datasets corresponding to individual NMR experiments. The API returns download URLs for raw data archives (ZIP files preserving the original directory structure of the spectrometer output), which were retrieved and extracted automatically. This programmatic access contrasts sharply with older repositories such as SDBS, which lack any public API and require manual or browser-automated retrieval. The raw FID data obtained from nmrXiv preserves the full time-domain signal, enabling custom reprocessing pipelines (apodization, Fourier transform, phase correction, baseline correction) rather than relying on pre-processed peak tables whose processing parameters are often undocumented. This level of data fidelity is critical for machine-learning applications where model performance depends on consistent and reproducible spectral preprocessing across the entire training corpus.

3.1.8. NMRBank

NMRBank is a large-scale experimental NMR database constructed through automated text mining of open-access PubMed Central articles using the NMRExtractor pipeline [129]. The pipeline processes the full-text corpus of 5,734,869 PMC articles in three stages. First, regular expressions identify paragraphs containing NMR data, yielding 380,220 candidate paragraphs from 58,795 articles. Second, a fine-tuned Mistral-7b large language model extracts structured fields from each paragraph, including compound IUPAC names, ^{1}H and ^{13}C NMR chemical

shifts, experimental conditions, and a confidence score derived from cumulative log-probabilities. The model was trained on 1,000 manually annotated NMR paragraphs and achieves greater than 0.85 extraction accuracy across diverse reporting formats. Third, IUPAC names are converted to canonical SMILES using ChemDraw with OPSIN as a fallback, followed by RDKit standardization. Entries with confidence scores above 0.8 exceed 97% accuracy.

The resulting dataset contains 225,809 entries, of which 156,621 were successfully converted to SMILES representations, corresponding to 149,135 unique molecular structures. The data are distributed as a CSV file with columns for SMILES, ^{1}H and ^{13}C chemical shifts as text strings, experimental conditions, confidence scores, and PMC accession numbers linking each entry to its source publication. In the present work, NMRBank serves as a supplementary NMR source after NMRexp and NMRShiftDB2. Chemical shifts are extracted by parsing floating-point values from the text field, providing individual peak positions in ppm. Although the text-mined shifts lack the multiplicity and coupling-constant annotations available in curated databases, NMRBank contributes 149,000 compounds that substantially expand chemical-space coverage, particularly for natural products and pharmaceutical intermediates whose NMR data appear only in the primary literature.

3.1.9. Specialized and Emerging Databases

Several smaller or more specialized databases were also evaluated for their potential contributions to the training corpus. Each serves a distinct niche in the NMR data ecosystem, and while none individually approaches the scale of NMRexp or NP-MRD, they collectively illustrate the diversity of data representations and curation philosophies in the field.

2DNMRGym is the first large-scale annotated experimental dataset for machine learning on two-dimensional NMR spectra, introduced by Li, Xu, and Hong [105]. The dataset contains 22,348 heteronuclear single quantum coherence (HSQC) spectra compiled from HMDB and the CH-NMR-NP natural products database, with molecular identifiers cross-referenced against PubChem. Each entry pairs an HSQC cross-peak table with the corresponding molecular graph and SMILES string, with peak positions stored as carbon-index-mapped chemical shift coordinates rather than raw spectral images. A held-out evaluation set of 479 spectra was manually annotated and cross-validated by three domain experts. Two-dimensional NMR techniques such as HSQC spread information across two frequency axes by correlating ^{1}H chemical shifts with those of ^{13}C, so that each cross peak encodes a direct one-bond coupling. This additional dimension dramatically reduces spectral congestion and provides atom-level connectivity information absent from 1D spectra.

PROSPRE (PROton Shift PREdictor) is a machine-learning web server that predicts ^{1}H NMR chemical shifts for small molecules in four common solvents: D_2O, $CDCl_3$, DMSO-d_6, and CD_3OD [130]. The underlying graph neural network was trained on 4,207 experimentally measured ^{1}H chemical shifts from 577 curated molecules drawn from HMDB (430 molecules), BMRB (103 molecules), and GISSMO (44 molecules). The predictor achieves a mean absolute error of 0.10 ppm in water and 0.19 ppm in chloroform. PROSPRE's downloadable training and holdout datasets offer a compact but carefully curated resource for benchmarking solvent-correction methods. For downstream applications such as structure–property regression from NMR descriptors, solvent-aware shift predictions allow spectra acquired under different conditions to be harmonized to a common reference frame before featurization.

The IR-NMR Multimodal Computational Spectra Dataset, published by Zipoli, Alberts, and Laino of IBM Research Europe, provides computationally generated infrared and NMR spectra for a large set of organic molecules extracted from patent literature [109]. The dataset contains IR spectra for 177,461 compounds alongside ^{1}H and ^{13}C chemical shifts for a subset of 1,255 compounds. Chemical shifts were computed using Car-Parrinello molecular dynamics at the DFT level with the PBE exchange-correlation functional and Goedecker-Teter-Hutter pseudopotentials, then referenced to tetramethylsilane. The NMR parquet file records frame-averaged chemical shifts, raw per-atom shielding tensors, and standard deviations across molecular dynamics frames, providing a uniquely detailed view of conformational averaging effects on chemical shifts.

The logD-NMR dataset, curated by Leniak, Pietrus, and Kurczab at Celon Pharma and the Polish Academy of Sciences [131], takes a fundamentally different approach to NMR data representation. Rather than storing raw spectra or peak lists, the dataset converts each compound's SMILES string to predicted ^{1}H and ^{13}C chemical shifts via the NMRShiftDB2 prediction engine, then histograms the resulting shift lists into 200-bin

vectors. The dataset contains over 1,200 pharmaceutical compounds with chromatographic hydrophobicity index log D labels measured at three pH conditions (0.26, 7.4, and 10.5). This bucketed pseudo-spectral representation directly demonstrates the use of NMR-derived descriptors as inputs to QSPR models, achieving RMSE values below 0.6 and $Q^2 > 0.7$ using 1D convolutional neural networks on the fused ^{1}H/^{13}C representation.

The Birmingham Metabolite Library (BML-NMR) was a publicly accessible database of 1D ^{1}H and 2D ^{1}H J-resolved NMR spectra of 208 authentic metabolite standards [132]. Each metabolite was prepared at three physiologically relevant pH values (6.6, 7.0, and 7.4), and sixteen NMR spectra were recorded per compound on a 500 MHz spectrometer, encompassing both 1D and 2D experiments with varying acquisition parameters. The resulting 3,328 spectra were stored in a purpose-built MySQL database compliant with Metabolomics Standards Initiative reporting requirements. This systematic variation of pH conditions was particularly valuable because chemical shift positions of ionizable metabolites are pH-dependent. However, the BML-NMR server has been unreachable since approximately September 2025, rendering the full database inaccessible. Despite recovery efforts, only 3 of the 3,328 spectra could be retrieved, representing a near-total loss of this public resource and underscoring the vulnerability of spectral databases hosted on single institutional servers without federated archiving.

3.1.10. PubChem as Identifier Resolution Service

PubChem, maintained by the National Center for Biotechnology Information (NCBI), is the world's largest open-access chemical database, containing records for over 116 million unique chemical compounds aggregated from more than 900 data sources [133]. Each unique chemical structure is assigned a Compound Identifier (CID), a stable integer that serves as a canonical reference. PubChem provides bulk data access through the NCBI FTP server, including the CID-SMILES.gz file (approximately 1.4 GB compressed), which maps every CID to its canonical SMILES string. Alongside SMILES, PubChem provides InChI representations developed by IUPAC and the InChI Trust, offering a layered, non-proprietary canonical identifier suitable for exact-match lookups and cross-database deduplication [134].

A persistent challenge in cheminformatics is identifier resolution: the same molecule may appear under different names, registry numbers, or structural representations across databases. HMDB, SDBS, and NMRShiftDB2 each use their own internal identifiers, and even ostensibly standard representations such as SMILES can vary between software implementations due to differences in canonicalization algorithms, stereochemistry handling, and aromaticity perception. PubChem addresses this problem by serving as a universal cross-reference hub, with its Identifier Exchange Service and bulk FTP data enabling mapping between CAS Registry Numbers, IUPAC names, InChI, InChIKey, and SMILES through the CID as intermediary. In this work, the CID-SMILES.gz file was used to resolve compound identifiers across heterogeneous NMR spectral databases into a unified canonical SMILES representation, which was subsequently re-canonicalized using RDKit to ensure consistency with the molecular representations used throughout the modeling pipeline [135]. This two-stage canonicalization—first through PubChem and then through RDKit—eliminates ambiguity arising from differing SMILES conventions and enables reliable structure-based joins across datasets that would otherwise lack a common identifier.

3.2. Thermophysical Property and Fuel-Specific Databases

The thermophysical property labels used throughout this work were assembled from six complementary sources that collectively span more than 1.2 million experimental data points across thousands of compounds. These sources differ in accuracy, chemical coverage, and data format, but together they provide sufficient breadth and depth to train machine learning models for viscosity, surface tension, and density prediction. Table 2 presents a consolidated overview of each source, including the number of compounds and data points contributed to the training pipeline.

Source	Compounds	Data Points	Properties
NIST ThermoML Archive	1,272 (pure visc.)	1,161,660	Viscosity, surface tension, density (pure + mixture)
`chemicals`/`thermo` Libraries	1,042 (visc.)	14,750	Viscosity, surface tension, density, T_b, T_f
CoolProp Reference Fluids	124	17,360	Viscosity, surface tension, density, C_p, k
NIST ILThermo	1,456	33,064	Viscosity, surface tension (ionic liquids)
NIST Chemistry WebBook	74	44,000	Viscosity, surface tension, density (saturation curve)
Joback Group Contribution	4,988	49,880	Viscosity, surface tension (estimated)
DDB / UNIFAC Free Tables	500	–	Group interaction parameters, Antoine coefficients

Table 2: Summary of thermophysical property sources. Data point counts include all temperatures and properties extracted from each source. The Joback estimates carry 10–30% uncertainty and were used for pretraining and data augmentation rather than as ground truth.

3.2.1. NIST ThermoML Archive

ThermoML is the IUPAC-standard XML schema for encoding experimentally determined thermodynamic and thermophysical property data, first formalized by Frenkel, Chirico, Diky, Yan, Dong, and Muzny in 2005 [136]. The schema captures compound identity (CAS Registry Number, InChI, InChIKey, molecular formula), experimental conditions (temperature, pressure, composition), property values with uncertainties, and metadata on measurement methodology. Each data file maps to a single journal article and encodes one or more pure-or-mixture data blocks, where properties are grouped by type (transport, surface, volumetric) and linked to constraint and variable definitions that specify whether temperature, pressure, or composition was scanned or held fixed. Standard SI units are enforced throughout: viscosity in Pa·s, density in kg/m^3, and surface tension in N/m. The NIST ThermoML archive, distributed as a 181 MB compressed tarball expanding to approximately 3.9 GB, contains 23,846 JSON-converted files derived from the original XML submissions to five journals from 2003 through 2019: the *Journal of Chemical and Engineering Data*, the *Journal of Chemical Thermodynamics*, *Fluid Phase Equilibria*, the *International Journal of Thermophysics*, and *Thermochimica Acta* [31]. Because the archive is a product of the U.S. federal government, it carries no copyright restrictions and is released into the public domain.

Parsing the 11,923 files that contain property data yielded 1,161,660 total data points distributed across six categories: pure-compound density (186,394 entries), pure-compound viscosity (44,440 entries from approximately 1,272 unique compounds), pure-compound surface tension (9,175 entries), mixture density (659,550 entries), mixture viscosity (223,807 entries), and mixture surface tension (38,294 entries). The viscosity data span a wide range of chemical classes, from light hydrocarbons and alcohols to heavy polyols and substituted aromatics, with temperatures typically between 273 K and 373 K and pressures at or near atmospheric. For the present work, the pure-compound viscosity and surface tension entries were the primary training targets, while the mixture data were reserved for future mixture-property modeling. Working with ThermoML data requires navigating several practical challenges. The JSON structure is deeply nested, with property types buried inside property group sub-dictionaries, and temperature and pressure may appear as either variable definitions (varying across data points within a block) or constraint definitions (fixed for the entire block), requiring the parser to merge both sources to reconstruct complete state conditions for each measurement. Compound identification relies on InChI strings for SMILES conversion, but a non-trivial fraction of entries contain salts, polymers, or ionic liquids whose InChI cannot be canonicalized to a valid SMILES, reducing the usable compound count. These parsing procedures were

implemented in a custom Python script that flattened the nested JSON structures, resolved property types by keyword matching, and exported the results as six CSV files indexed by InChI and InChIKey [30], [137].

3.2.2. Open-Source Correlation Libraries

The `chemicals` and `thermo` open-source Python libraries, both components of the Chemical Engineering Design Library (ChEDL) developed by Caleb Bell, served as the primary source of computed thermophysical properties for compounds lacking direct experimental measurements [32], [138]. The `chemicals` library provides low-level access to pure-component property correlations drawn from standard references including Perry's Chemical Engineers' Handbook [139], the VDI Heat Atlas [140], and the DIPPR evaluated-data compilations. For liquid viscosity, the library implements several temperature-dependent models: DIPPR Equation 101 (a four-parameter exponential form covering 337 compounds from Perry's Table 2-313), three variants of the Viswanath–Natarajan correlation (432 compounds in the three-term form), the Dutt–Prasad three-term correlation (100 compounds), and VDI PPDS polynomial fits (271 compounds). For surface tension, available correlations include the Jasper–Lange linear model (522 compounds), VDI PPDS Series 11 coefficients (272 compounds), and the Mulero–Cachadina formulation (115 compounds). Across all correlation sources, approximately 1,042 unique compounds have at least one viscosity correlation available, and approximately 690 compounds have surface tension data. The higher-level `thermo` library wraps these correlations with automatic method selection and ranking, choosing the best available data source for each compound at the requested temperature and pressure. Using these libraries, viscosity and surface tension values were evaluated at ten temperatures spanning each compound's liquid range, producing a computed-property dataset of approximately 7,860 viscosity and 6,890 surface tension evaluations.

For a smaller set of high-accuracy reference fluids, properties were obtained from CoolProp, an open-source C++ library with Python bindings that implements multiparameter Helmholtz energy equations of state [141]. CoolProp covers 124 pure and pseudo-pure fluids whose thermodynamic and transport properties are computed from reference-quality correlations validated against primary experimental data. Because Helmholtz-based formulations provide internally consistent thermodynamic surfaces, CoolProp yields lower uncertainty than empirical fits for the fluids it supports, typically sub-percent for density and 1–3% for viscosity and surface tension along the saturation curve. In this work, CoolProp contributed viscosity, surface tension, density, thermal conductivity, heat capacity, vapor pressure, and enthalpy of vaporization for all 124 fluids at 20 temperatures each, yielding approximately 17,360 property evaluations that serve as high-fidelity anchor points for model calibration. Together, these three libraries—`chemicals`, `thermo`, and CoolProp—enabled the construction of a computed-property dataset spanning over 1,000 compounds without requiring access to proprietary databases such as DIPPR 801. All property evaluations are deterministic and reproducible given a compound identifier (CAS number or SMILES) and the library version.

3.2.3. NIST ILThermo

The NIST Standard Reference Database 147, known as the Ionic Liquids Thermodynamics Database or ILThermo, is a critically evaluated repository of experimental thermophysical property data for ionic liquids maintained by the Thermodynamics Research Center at the National Institute of Standards and Technology [142], [143]. Ionic liquids—salts that are liquid at or near room temperature, typically composed of a bulky organic cation paired with an inorganic or organic anion—exhibit negligible vapor pressure, wide liquid ranges, high thermal stability, and tunable physicochemical properties that have attracted intense research interest in applications ranging from electrochemistry and catalysis to carbon capture and lubrication [144]. ILThermo version 2.0 provides a REST API that returns structured JSON data for programmatic queries by property type, chemical identity, author, or keyword. In this work, the API was queried for all single-component viscosity and surface tension datasets, yielding 1,456 unique ionic liquid compounds and 33,064 data points in total: 1,345 compounds with 26,189 viscosity measurements and 506 compounds with 6,875 surface tension measurements. Each dataset was fetched individually with a 0.3-second delay between requests to avoid overloading the server, and the returned values were converted to common engineering units (mPa·s for viscosity, mN/m for surface tension).

Ionic liquid data from ILThermo serve a distinct and complementary role relative to the organic compound datasets that form the bulk of the training corpus. Conventional organic liquids in the ThermoML and `chemicals` databases are predominantly small neutral molecules—hydrocarbons, alcohols, esters, and ethers with molecular

weights below 300 g/mol—whose viscosities at room temperature typically fall between 0.3 and 10 mPa·s and whose surface tensions range between 15 and 40 mN/m. Ionic liquids, by contrast, exhibit viscosities that are one to three orders of magnitude higher, commonly 30 to 500 mPa·s at 298 K with some exceeding 10,000 mPa·s, and surface tensions that cluster between 25 and 50 mN/m owing to their strong Coulombic cohesion. Including ionic liquids in the training set therefore extends the dynamic range of the viscosity target by roughly two orders of magnitude, which improves model extrapolation to highly viscous organic liquids and fuel blends at low temperatures. Furthermore, ionic liquids introduce structural motifs—imidazolium, pyridinium, pyrrolidinium, and phosphonium cations paired with fluorinated or cyano anions—that are absent from conventional organic datasets, encouraging the model to learn more generalizable structure–property relationships rather than overfitting to the narrow chemical space of light hydrocarbons.

3.2.4. NIST Chemistry WebBook and CoolProp Reference Fluids

The NIST Chemistry WebBook (Standard Reference Database 69) provides thermophysical property data for 74 reference fluids computed from validated Helmholtz-energy equations of state, representing the highest-fidelity data in this collection [145]. For each fluid, the WebBook supplies liquid-phase viscosity, surface tension, and density along the saturation curve at 1 K increments from the triple point to the critical point, yielding approximately 600 data points per compound and roughly 44,000 data points in aggregate across the 74 fluids. Although small in chemical scope—the 74 fluids are predominantly light hydrocarbons, refrigerants, and common industrial solvents—these data carry sub-percent uncertainties for density and 1–3% uncertainties for viscosity and surface tension, making them suitable as ground-truth benchmarks for evaluating model accuracy. The WebBook data were scraped from the NIST Fluid Properties interface using automated HTTP requests, and each fluid's saturation-curve data were exported as a CSV file with columns for temperature, viscosity, surface tension, and density. A master index consolidating all 74 fluids was compiled for rapid cross-referencing.

3.2.5. Dortmund Data Bank Free Tables

The Dortmund Data Bank (DDB), maintained by DDBST GmbH, is one of the world's largest collections of thermodynamic and transport property data for pure compounds and mixtures [146]. Although the full DDB is a commercial product, DDBST publishes a subset of group-interaction parameters and reference data on its public website. In this work, three categories of free DDB data were incorporated: UNIFAC group-interaction parameters for activity-coefficient estimation, Antoine vapor-pressure coefficients for several hundred compounds, and phase-change enthalpies. The UNIFAC parameters are particularly valuable because they enable the UNIFAC-VISCO mixture viscosity method [43], which decomposes mixture transport properties into combinatorial and residual contributions using the same functional-group framework employed in vapor–liquid equilibrium calculations [42]. Additionally, Antoine coefficients from the DDB free tables were used for vapor-pressure calculations needed in corresponding-states viscosity and surface tension correlations. Six supplementary files scraped from the NIST Chemistry WebBook provided Antoine parameters, phase-change data, and thermochemical properties for 94 common C_5–C_{20} hydrocarbons relevant to jet fuel applications.

3.2.6. Group Contribution Estimates

To extend property coverage beyond the approximately 1,300 compounds for which experimental or high-quality correlation data were available, group contribution estimates were computed for 4,988 compounds sampled from the NMRexp SMILES pool using the Joback method [1]. The Joback method estimates critical temperature, critical pressure, and normal boiling point from additive contributions of 41 first-order molecular substructures identified by SMARTS pattern matching in RDKit [147]. These critical properties were then fed into corresponding-states correlations: the Letsou–Stiel method for viscosity and the Brock–Bird method for surface tension [3]. Property values were generated at five temperatures (250, 300, 350, 400, and 450 K), yielding approximately 24,940 viscosity estimates and 24,940 surface tension estimates. These group contribution values carry typical errors of 10–30% for hydrocarbons and larger errors for polar or multifunctional compounds, so they were used exclusively for model pretraining and data augmentation rather than as ground-truth labels. The pretraining strategy exposes the model to a broader chemical space during initial weight optimization, after which the model is fine-tuned on the smaller but more accurate experimental datasets from ThermoML and the `chemicals` library.

Separately, the `chemicals` library was used to generate expanded property evaluations for its approximately 786 viscosity compounds and 689 surface tension compounds at ten temperatures each, using the best available correlation for each compound as selected by the `thermo` ranking algorithm. Density data were generated for approximately 441 compounds using Perry's DIPPR Equation 105 and VDI PPDS polynomial correlations, producing 8,718 density data points that were supplemented by the 44,000 NIST WebBook density values to create a combined density training set spanning roughly 500 compounds. Thermal conductivity (101,333 data points across 9,861 compounds) and heat capacity (101,361 data points across 9,861 compounds) were also generated as multi-task learning targets, providing auxiliary regression objectives that share molecular representations with the primary viscosity and surface tension tasks.

3.2.7. Identifier Reconciliation

Merging data across these six sources presented a persistent identifier-reconciliation challenge that warrants explicit discussion. The NIST Chemistry WebBook uses internal alphanumeric codes, ThermoML records carry CAS Registry Numbers and InChI strings, the `chemicals` library indexes primarily by CAS number, ILThermo uses compound names and molecular formulas, and the NMR spectral databases (NMRexp and NMRShiftDB2, described in the preceding section) use SMILES as their primary key. Linking records across sources required converting between CAS, InChI, InChIKey, and canonical SMILES through a combination of PubChem API lookups, the `chemicals.identifiers.search_chemical` function, and RDKit canonicalization [133]. This process introduced occasional mismatches for stereoisomers (where CAS distinguishes *cis* and *trans* forms but canonical SMILES may not), for salts and hydrates (where InChI encodes the complete stoichiometry but SMILES typically represents only the organic component), and for compounds absent from PubChem. To mitigate these errors, all merged datasets were filtered by molecular-weight consistency checks—rejecting any cross-reference where the molecular weights implied by the two identifiers differed by more than 1 g/mol—and, where possible, validated against the original compound names stored in the source databases. The final unified compound index, stored as a master CSV file mapping CAS, InChI, InChIKey, SMILES, name, and molecular formula, contains 1,225 compounds for which all identifier types were successfully reconciled and at least one thermophysical property and one molecular representation (fingerprint, descriptor, or NMR spectrum) are available.

3.3. Fuel-Specific Data

3.3.1. AFRL Jet Fuel Property Compilation

The primary source of fuel-mixture ground truth in this work is the AFRL technical report AFRL-RQ-WP-TR-2020-0017, authored by Edwards and published in January 2020 [119]. This 136-page document constitutes the most comprehensive publicly available compilation of thermophysical properties for both conventional petroleum-derived and alternative jet fuels. Edwards catalogues properties for three Category A reference fuels—JP-8 (POSF 10264), Jet-A (POSF 10325), and JP-5 (POSF 10289)—alongside more than ten alternative fuels, including Shell SPK (POSF 5729), Sasol IPK (POSF 7629), Syntroleum S-8 (POSF 5018), UOP camelina HEFA (POSF 10301), Gevo ATJ (POSF 11498), and several others. Each fuel is identified by its POSF (Properties of Fuels) tracking number, a unique integer assigned by the AFRL Fuels and Energy Branch to every fuel sample that enters the characterization pipeline.

The report is distributed as thirteen CSV files, each corresponding to a different property or property class. Tabulated quantities include density, dynamic viscosity, surface tension, speed of sound, bulk modulus, lubricity, distillation curves, derived cetane number, flash point, freeze point, net heat of combustion, hydrogen content, and molecular weight distributions. Temperature-dependent data are provided where available: density–temperature linear equations derived from the CRC Handbook and the Department of Defense World Fuel Survey, viscosity at $-20°C$ and $-40°C$, and surface tension as a function of temperature with linear-fit coefficients. Detailed GC×GC hydrocarbon-class composition breakdowns enable composition-weighted property modeling of real fuel mixtures. Of particular value to this work are the low-temperature viscosity measurements for alternative fuels, the surface tension linear regressions, and the GC×GC composition data that supply the mole fractions needed to aggregate pure-compound predictions into fuel-level estimates.

Six fuels from the Edwards compilation serve as the validation set throughout Phases 3 and 4 of this study. Table 3 summarizes these fuels together with the POSF identifiers and measured property ranges used for model evaluation.

Fuel	POSF	Type	Primary Use
JP-5	10289	Petroleum	Navy / carrier ops
Jet-A	10325	Petroleum	Commercial aviation
HRJ-Camelina	7720	HEFA	Bio-derived SAF
Shell SPK	5729	Fischer–Tropsch	Synthetic paraffinic
Sasol IPK	7629	Fischer–Tropsch	Iso-paraffinic
Gevo ATJ	10151	Alcohol-to-jet	Bio-derived SAF

Table 3: Six validation fuels drawn from the AFRL compilation [119], spanning petroleum baselines and four distinct alternative-fuel production pathways.

These six fuels span the compositional diversity of the current jet fuel landscape. The two petroleum-derived fuels, JP-5 and Jet-A, are complex mixtures of hundreds of hydrocarbons across the C_8–C_{16} range, containing normal and branched paraffins, cycloparaffins, and aromatics. The four alternative fuels, by contrast, are enriched in specific hydrocarbon classes depending on their synthesis pathway: the Fischer–Tropsch fuels (Shell SPK and Sasol IPK) are predominantly iso-paraffinic with negligible aromatic content, the HEFA fuel (HRJ-Camelina) retains some cycloparaffinic character from its bio-feedstock, and the alcohol-to-jet fuel (Gevo ATJ) is a narrow-cut iso-paraffinic kerosene. This diversity provides a stringent test of whether models trained on pure-compound data can generalize to fuel mixtures whose compositions differ markedly from one another.

3.3.2. Burnett Reference Implementation

The modeling architecture and featurization strategy in this work build upon a reference implementation by Burnett, released as an open-source repository under the GNU General Public License v3 [148]. Burnett provides two parallel neural network pipelines for predicting the temperature-dependent viscosity of pure hydrocarbons. The first pipeline takes simulated proton NMR spectra as input: SMILES strings for each training compound are submitted to the NMReDB spectral prediction service, which returns chemical shift positions and hydrogen multiplicities that are convolved with Lorentzian line shapes to produce a continuous ^{1}H spectrum discretized at 6554 points. These spectra are fed into a four-layer one-dimensional convolutional neural network with 32, 16, 8, and 4 filters, followed by three fully connected layers (128, 64, and 2 neurons), which outputs two Arrhenius-type coefficients A and B such that $\ln \eta = A + B/T$.

The second pipeline replaces the NMR spectrum with a seven-element functional group vector counting CH_3, CH_2, CH, quaternary C, aromatic ring, cyclic CH_2, and long-chain CH_2 groups, identified by SMARTS substructure matching through RDKit. This vector is processed by a compact three-layer fully connected network (12, 4, and 2 neurons). Both pipelines are trained on approximately 65 pure hydrocarbons whose experimental viscosity–temperature data are drawn from the NIST ThermoData Engine via ThermoML-format equations. The training set spans normal alkanes, branched alkanes, cycloalkanes, and aromatics representative of jet fuel constituents, providing coverage of the major hydrocarbon families present in the AFRL fuels. Burnett further demonstrates fuel-mixture prediction by applying the trained models to JP-5 (POSF 10289), Jet-A (POSF 10325), and HRJ-Camelina (POSF 7720) using composition-weighted averages of constituent pure-compound predictions. This composition-weighted aggregation strategy is adopted directly in the present work and extended to surface tension and density.

3.3.3. NMR-Based Fuel Property Prediction Literature

Two independent lines of published research establish the precedent that NMR spectra encode sufficient chemical information for quantitative fuel property prediction. Abdul Jameel et al. demonstrated that seven functional group abundances extracted from ^{1}H NMR spectra—paraffinic CH_3, paraffinic CH_2, paraffinic CH, olefinic $CH{=}CH_2$, naphthenic $CH{-}CH_2$, aromatic C–CH, and ethanolic OH—together with molecular weight and a

branching index, suffice to predict research and motor octane numbers with $R^2 = 0.99$ and mean absolute errors of approximately 1.2 octane numbers [19]. That study compiled data for 128 pure hydrocarbons, 123 hydrocarbon–ethanol blends of known composition, and 30 FACE (Fuels for Advanced Combustion Engines) gasoline–ethanol blends. In a subsequent paper, Abdul Jameel formalized the spectral integration procedure by defining precise chemical shift windows for six hydrocarbon functional groups and introduced a branching index parameter that captures the degree of methyl substitution on the carbon backbone [114]. Applied to a dataset of 281 fuels comprising pure compounds and complex blends, this methodology codified a reproducible, standardized protocol for translating ^{1}H NMR spectra of complex fuel mixtures into a compact functional group representation suitable for machine learning.

Parker et al. extended NMR-based fuel property prediction from combustion metrics to thermophysical properties directly relevant to sustainable aviation fuel certification [21]. Using ^{1}H–^{13}C HSQC two-dimensional NMR spectroscopy, the authors identified and quantified discrete atom types—a finer-grained decomposition than the functional groups used by Abdul Jameel—by integrating peaks on the ^{1}H spectrum with carbon-type assignments from the HSQC dimension. Multilinear regression models built on these atom-type group contributions predicted liquid density within 0.00–5.59% absolute error and surface tension within 0.29–4.41% absolute error across six real aviation fuels. The approach is designed to support prescreening under ASTM D4054, where candidate SAFs must demonstrate acceptable thermophysical properties before proceeding to costly engine testing. By requiring only milligram-scale fuel samples and a single NMR measurement, the method offers a rapid, low-cost alternative to conventional ASTM property tests.

Collectively, the progression from Abdul Jameel's octane-number models through Parker's density and surface tension predictions demonstrates that NMR spectroscopy provides a general-purpose analytical framework for fuel property prediction. The same spectral measurement can be reprocessed to target different properties as modeling objectives change. The present work builds on this foundation by replacing multilinear regression with deep learning architectures (Phases 1–2), extending the target properties to include temperature-dependent viscosity, and introducing a composition-weighted aggregation scheme (Phase 3) that enables fuel-mixture prediction from pure-compound models without requiring mixture-level training data.

3.4. Feature Engineering and Data Quality

The raw data sources described in the preceding sections span heterogeneous formats, naming conventions, and levels of completeness. Before any modeling can proceed, each compound must be reduced to a fixed-length numerical representation that captures the relevant chemistry while remaining compatible with standard machine learning algorithms. This section describes the feature engineering pipelines for NMR spectra and molecular fingerprints, the identifier resolution challenges that arise when linking records across databases, and the data quality issues that constrain the effective training set size.

3.4.1. NMR Feature Extraction Pipeline

Experimental ^{1}H NMR data in the NMRexp database arrive as structured peak lists rather than continuous spectra. Each entry is a list of tuples recording the multiplicity, coupling constants, integral (in hydrogen equivalents), and the high and low bounds of the chemical shift range for every resolved signal [13]. A triplet at 3.68 ppm integrating for two hydrogens with a coupling constant of 3.8 Hz, for example, is stored as `('t', ['3.8Hz'], '2H', 3.68, 3.68)`. Converting these discrete descriptors into a uniform numerical representation requires two stages: spectrum reconstruction via lineshape simulation, and domain-informed feature extraction from the reconstructed spectrum.

In the first stage, the peak list is parsed and each signal is rendered onto a 1200-point grid spanning 0–12 ppm. Singlets, doublets, triplets, and quartets are simulated as sums of Lorentzian lineshapes with Pascal's triangle intensity ratios, using the reported coupling constants to determine line separations in ppm (dividing Hz values by an assumed spectrometer frequency of 400 MHz). Higher-order multiplets such as doublets of doublets are generated by successive splitting: each existing line is split into a doublet about its center for every additional coupling constant. Unresolved multiplets, denoted "m" in the peak list, are rendered as Gaussian envelopes whose width matches the reported chemical shift range. Each Lorentzian line has a half-width at half-maximum of 0.01

ppm, chosen to be narrow enough to preserve peak identity while broad enough that the 0.01 ppm grid spacing captures the lineshape adequately. After all peaks have been rendered, the spectrum is max-normalized to the interval $[0, 1]$.

The 1200-point spectrum is too high-dimensional relative to the available training set (hundreds to low thousands of compounds), so a second stage extracts 120 domain-informed features organized into four groups.

The first group consists of 60 fine spectral bin features. The 0–12 ppm range is divided into 30 bins of 0.4 ppm width, and for each bin the mean and maximum signal intensity are recorded. These capture the overall shape of the spectrum at a resolution matched to the precision of the chemical shift data.

The second group consists of 27 chemical shift region features. Nine chemically meaningful regions of the ^{1}H spectrum are defined, and for each region three quantities are computed: the integral (sum of signal, proportional to the number of protons in that environment), the fraction of the total integral attributable to that region, and the number of distinct peaks (counted as upward threshold crossings at 5% of maximum intensity). Table 4 lists the nine regions and their chemical significance. The aromatic and olefinic regions are particularly informative for fuel applications because aromatic content correlates strongly with density and surface tension, while the alkyl regions encode chain length information relevant to viscosity [16], [22].

Region	**Range (ppm)**	**Chemical significance**
Alkyl high-field	0.0–1.0	Methyl groups, shielded C–H
Alkyl mid	1.0–2.0	Methylene, methine
Alkyl α-hetero	2.0–3.0	C–H adjacent to O, N, halogen
Oxygenated	3.0–4.5	O–CH, N–CH environments
Olefinic low	4.5–5.5	Vinyl, allyl
Olefinic high	5.5–6.5	Vinyl
Aromatic low	6.5–7.5	Aromatic C–H
Aromatic high	7.5–8.5	Downfield aromatic, heteroaromatic
Aldehyde/downfield	8.5–12.0	Aldehyde, carboxylic acid

Table 4: Chemical shift regions used for ^{1}H NMR feature extraction. Boundaries follow standard assignment tables [5], [6].

The third group consists of 8 global spectral statistics. These are the spectral centroid (intensity-weighted mean chemical shift), spread (intensity-weighted standard deviation), skewness, kurtosis, total peak count, Shannon entropy of the normalized spectrum, the fraction of the grid with signal above the noise floor, and the maximum peak height. The centroid and spread summarize the overall position and breadth of the hydrogen distribution: a long-chain alkane has a centroid near 1.3 ppm with small spread, while a polycyclic aromatic has a centroid above 7 ppm.

The fourth group consists of 25 peak-list features extracted directly from the parsed peak data before spectrum reconstruction. These features preserve absolute hydrogen count information that is lost during max-normalization of the continuous spectrum. Total hydrogen count is a strong proxy for molecular weight, which is in turn a dominant predictor of viscosity. The peak-list features include: 3 size proxies (total hydrogen count, number of peaks, and maximum single-peak integral); 9 absolute hydrogen counts per chemical shift region; 7 multiplicity distribution fractions (singlet, doublet, triplet, quartet, multiplet, complex, and other); 3 coupling constant statistics (mean and maximum J value, and the fraction of peaks reporting coupling information); and 3 derived ratios (alkyl hydrogen fraction, aromatic hydrogen fraction, and a CH_2/CH_3 proxy computed as the ratio of 1.0–2.0 ppm hydrogen count to 0.0–2.0 ppm hydrogen count). The CH_2/CH_3 ratio increases with chain length and branching degree, providing a direct structural descriptor for the aliphatic hydrocarbons that dominate jet fuel composition.

The combined 120-dimensional feature vector (95 spectral + 25 peak-list) can be computed for any compound that has a parsed ^{1}H NMR peak list, requiring only a few milliseconds per compound and no GPU resources.

3.4.2. Molecular Fingerprints and Structural Descriptors

Phase 1 of this work uses molecular fingerprints as the sole input representation, providing a structure-only baseline against which NMR-based models are compared. Morgan circular fingerprints of radius 2, folded to 2048 bits, are computed from canonical SMILES using RDKit [55], [147]. These extended-connectivity fingerprints encode the presence or absence of circular substructures up to four bonds in diameter, capturing functional groups, ring systems, and local connectivity patterns [54]. The radius-2, 2048-bit configuration is a widely adopted default that balances expressiveness against sparsity and has been shown to perform well across diverse property prediction tasks [55].

In addition to binary fingerprints, Phase 1 models use a set of RDKit molecular descriptors including molecular weight, heavy atom count, number of rotatable bonds, topological polar surface area, and partition coefficient (log P). These continuous descriptors supplement the binary fingerprint by encoding global molecular properties that are not directly captured by substructure counts. Molecular weight and rotatable bond count are particularly relevant because viscosity scales approximately with molecular size and chain flexibility [45].

3.4.3. Identifier Resolution and Cross-Database Linking

Linking NMR spectra to thermophysical property measurements requires matching compound identities across databases that use different naming conventions, identifier systems, and canonicalization rules. NMRexp records compounds by SMILES extracted from journal articles via text mining [13]; NIST ThermoML uses CAS registry numbers and IUPAC names [136]; and DIPPR stores records by a proprietary compound index. Resolving these identifiers to a common canonical form is a prerequisite for constructing training sets.

SMILES canonicalization is the primary unification mechanism because SMILES strings encode molecular structure unambiguously (up to stereochemistry) and can be compared directly after canonicalization [134]. However, several systematic challenges arise in practice. Different databases may represent the same compound with or without explicit stereochemistry: ethanol appears as `CCO` in one source and `C(C)O` in another, both of which canonicalize identically, but (R)- and (S)-limonene may appear as the single achiral SMILES `CC(=CC1CC=C(C)C1)` when stereochemistry is not recorded. For fuel-relevant hydrocarbons, which are predominantly achiral, this ambiguity is minor, but for oxygenated or nitrogen-containing compounds it can lead to false matches or missed links.

A more consequential challenge is that text-mined SMILES in NMRexp contain errors at an estimated rate of 1–3%, including truncated strings, mismatched parentheses, and incorrect atom valences [129]. These invalid SMILES are detected and filtered during preprocessing by attempting to parse each string with RDKit and discarding any that fail sanitization. Additionally, NMR peak lists extracted from different papers for the same compound may disagree in integral values, coupling constants, or even the number of reported peaks, reflecting differences in solvent, concentration, temperature, or the original author's peak-picking conventions.

3.4.4. Data Quality and Reporting Variability

Several categories of data quality issues affect the NMR and property datasets used in this work.

Integral rounding is pervasive in literature-reported NMR data. Hydrogen integrals are almost always reported as integers (1H, 2H, 3H), even when the true integral is non-integer due to partial deuterium exchange, relaxation effects, or overlapping peaks. This quantization introduces noise in the peak-list features, particularly the total hydrogen count and the region-specific hydrogen counts.

Variable reporting depth is another concern. Some journal articles report full multiplicity and coupling constant information for every peak, while others report only chemical shifts and integrals, listing all peaks as singlets or multiplets. The fraction of peaks with coupling constant information varies from 0% to 100% across NMRexp entries, which creates missing-data patterns in the multiplicity and coupling features. The feature extraction pipeline handles this gracefully by computing multiplicity fractions relative to the number of peaks present and defaulting coupling statistics to zero when no coupling information is available.

Solvent and reference standard variations affect chemical shift values. Although most ^{1}H NMR spectra in NMRexp are referenced to tetramethylsilane (TMS) at 0.0 ppm, spectra acquired in dimethyl sulfoxide use the residual solvent peak at 2.50 ppm, and spectra in deuterated water use the HDO peak at 4.79 ppm. Systematic offsets of 0.01–0.05 ppm from imprecise referencing are common and are absorbed by the 0.4 ppm bin width in the feature extraction pipeline. Larger offsets from incorrect solvent referencing, though rare, can misassign peaks to adjacent chemical shift regions.

Property data quality in the thermophysical databases is generally higher than in the NMR databases because measurements follow standardized protocols (e.g., ASTM D445 for viscosity). Nevertheless, inter-laboratory reproducibility for viscosity measurements is typically 1–3%, and for surface tension 0.5–2%, which sets a floor on achievable prediction accuracy [27], [140].

3.4.5. Synthetic NMR Generation from Molecular Structure

The most severe data limitation is the small overlap between compounds with NMR spectra and compounds with thermophysical property measurements. Of approximately 1.48 million unique compounds in NMRexp, and approximately 2,238 compounds with viscosity measurements in the combined NIST/DIPPR databases, only 1,330 compounds appear in both datasets. For surface tension the overlap is somewhat larger, but the fundamental constraint remains: experimental NMR data exist overwhelmingly for compounds of interest to synthetic organic chemistry, while thermophysical property data exist overwhelmingly for industrially important solvents, fuels, and commodity chemicals. The intersection of these two populations is small.

To expand the effective training set, Phase 4 introduces a synthetic NMR generation pipeline that predicts approximate ^{1}H chemical shifts from SMILES alone. The approach uses a substructure-based lookup table derived from standard chemical shift correlation charts [5], [6]. For each hydrogen atom in the molecule, the algorithm applies a priority-ordered list of SMARTS patterns: aldehyde C–H at 9.7 ppm, aromatic C–H at 7.2 ppm, vinyl C–H at 5.0–5.3 ppm, O–CH at 3.3–3.7 ppm depending on hybridization, N–CH at 2.3–2.7 ppm, carbonyl α-CH at 2.1–2.3 ppm, benzylic CH at 2.3–2.5 ppm, generic methylene at 1.3 ppm, and generic methyl at 0.9 ppm. The first matching pattern determines the predicted shift for each hydrogen, and hydrogens with shifts within 0.15 ppm of each other are grouped into a single peak with an integral equal to the number of equivalent hydrogens. The predicted shifts have an estimated accuracy of 0.5–1.0 ppm, which is coarse by spectroscopic standards but adequate for the 0.4 ppm binning used in feature extraction.

The synthetic peaks are processed through the same Lorentzian broadening and feature extraction pipeline as experimental data, producing the identical 120-dimensional feature vector. Because all hydrogen atoms are accounted for, the total hydrogen count feature is exact (matching the molecular formula), though the distribution across chemical shift regions is approximate. Small Gaussian jitter (standard deviation 0.05 ppm) is added to predicted shifts to avoid artificially sharp features that would not appear in experimental spectra. This synthetic NMR pipeline enables NMR-like features to be computed for any compound with a valid SMILES string, expanding the potential training set from 1,330 to more than 6,000 viscosity compounds, at the cost of introducing systematic approximation error in the spectral features. The trade-off between data quantity and feature fidelity is evaluated empirically in Section 4.3.

4. Model Designs

4.1. Phase 1: Molecular Fingerprint Baselines

The objective of Phase 1 is to establish baseline predictive accuracy for viscosity and surface tension using only molecular structure, without any NMR spectral information. These baselines quantify how well standard cheminformatics representations capture thermophysical behavior and provide a reference against which subsequent NMR-augmented models (Phases 2–4) are compared. All data sources, feature generation procedures, and quality filters described in Section 3 apply to this phase.

4.1.1. Input Feature Representation

Each compound is represented by the concatenation of three complementary structural descriptors. First, 2048-bit Morgan circular fingerprints [55] computed at radius 2 encode the presence or absence of substructural neighborhoods up to four bonds from each atom, providing a hashed topological description that captures functional-group context. Second, 166-bit MACCS structural keys [57] record the presence of predefined pharmacophore-like fragments, including specific ring systems, heteroatom patterns, and chain lengths. Third, a set of RDKit molecular descriptors [147] computed by the RDKit toolkit supply continuous physicochemical quantities such as molecular weight, Wildman–Crippen $\log P$, topological polar surface area, Balaban's J index, and various partial-charge and surface-area partitions. After removing zero-variance columns, the combined feature vector has approximately 1,696 dimensions per compound.

Temperature enters the model through three engineered features: the absolute temperature T in kelvin, its reciprocal $1/T$ (motivated by the Arrhenius form of Equation 1), and the reduced temperature $T/1000$. These three scalar features are concatenated with the molecular embedding at the prediction head, enabling the model to learn temperature-dependent property surfaces for each compound.

4.1.2. Traditional Machine Learning Models

Five model families were compared in a compound-level five-fold cross-validation protocol using the Arrhenius parameterization. In this formulation, each model predicts the two Arrhenius coefficients A and B of Equation 1 from the molecular feature vector, and reconstructed viscosity curves are evaluated against held-out experimental data. Hyperparameters were tuned by randomized search with an inner three-fold cross-validation loop on the training partition of each outer fold. The five model families are XGBoost [33], Random Forest [34], LightGBM [65], Ridge regression, and Lasso regression.

Table 5 and Table 6 report the Arrhenius coefficient R^2 and the reconstructed-property R^2 for viscosity and surface tension, respectively. For viscosity, XGBoost achieves the highest reconstructed R^2 of 0.274, with Random Forest close behind at 0.308. The linear models (Ridge and Lasso) produce more stable but lower reconstructed performance ($R^2 \approx 0.11$– 0.13). All models display high variance across folds, reflecting the difficulty of predicting Arrhenius intercepts and slopes from fingerprints alone. For surface tension, XGBoost again leads with a reconstructed R^2 of 0.761, substantially higher than the viscosity case, suggesting that surface tension is more directly encoded in molecular topology.

Model	$R^2(A)$	$R^2(B)$	R^2(recon)	**Time (s)**
XGBoost	0.305	0.629	0.274	593
LightGBM	0.269	0.641	0.211	350
Random Forest	0.327	0.622	0.308	191
Ridge	0.332	0.536	0.132	0.5
Lasso	0.409	0.592	0.106	24

Table 5: Arrhenius-parameterized viscosity prediction: five-fold compound-level CV. $R^2(A)$ and $R^2(B)$ are the coefficient-of-determination for the Arrhenius intercept and slope, respectively. R^2(recon) evaluates reconstructed $\ln \eta$ against experimental values.

Model	$R^2(A)$	$R^2(B)$	R^2(recon)	**Time (s)**
XGBoost	0.564	0.456	0.761	1119
LightGBM	0.560	0.468	0.748	892
Random Forest	0.548	0.442	0.728	92
Ridge	0.445	0.428	0.622	0.9
Lasso	0.478	0.438	0.621	82

Table 6: Arrhenius-parameterized surface tension prediction: five-fold compound-level CV, same protocol as Table 5.

4.1.3. Neural Network Architecture

To move beyond the two-coefficient Arrhenius parameterization, a neural network was trained to predict $\ln \eta$ (viscosity) or surface tension directly, with temperature supplied as an explicit input alongside the molecular features. This direct formulation avoids forcing the model into a two-parameter functional form and allows the network to learn nonlinear temperature dependencies that deviate from ideal Arrhenius behavior.

The architecture follows a shared-encoder, property-specific-head design. The shared encoder is a stack of three fully connected layers with dimensions $d_{\text{in}} \to 512 \to 256 \to 128$, where d_{in} is the number of selected molecular features. Each layer applies a linear transformation, batch normalization, ReLU activation, and dropout ($p = 0.3$) in sequence, with Kaiming normal initialization for all weight matrices. The encoder output is a 128-dimensional molecular embedding. The three temperature features described in Section 4.1.1 are concatenated with this embedding to form a 131-dimensional vector, which is passed to a property-specific prediction head. Each head consists of a single hidden layer of 32 units (batch normalization, ReLU, dropout $p = 0.2$) followed by a linear output producing a scalar prediction.

Because Lasso feature selection is applied per fold to the full 1,696-dimensional feature vector, the effective input dimension varies between 22 and 38 features across folds, with a median of approximately 28. The Lasso step (regularization strength $\alpha = 0.1$) fits a linear model to the per-compound mean property value and retains only those features whose coefficients exceed 10^{-6} in absolute value. This aggressive dimensionality reduction is intentional: it prevents the network from memorizing bit-level fingerprint noise and forces reliance on the most informative descriptors.

4.1.4. Training Protocol

Training proceeds under a five-fold compound-level cross-validation protocol. Compound-level splitting ensures that all temperature-dependent measurements for a given molecule appear in exactly one partition, preventing data leakage between training and test sets. Within each training fold, 10% of the data is held out as a validation set for early stopping.

The loss function is the Huber loss with $\delta = 1.0$, which interpolates between mean squared error for small residuals and mean absolute error for large residuals. This choice mitigates the influence of outliers in the property databases without discarding them entirely. The optimizer is AdamW with a learning rate of 3×10^{-4} and weight decay of 10^{-4}. A cosine-annealing learning rate schedule with a 10-epoch linear warmup ramps the learning rate from near zero to its peak, then decays it smoothly to 10^{-6} over the remaining epochs. Gradient norms are clipped at 1.0 to stabilize training. Training runs for up to 500 epochs with early stopping triggered after 50 epochs of no improvement on the validation loss. Input features and targets are standardized to zero mean and unit variance using statistics computed on the training fold.

An ensemble of five independently initialized networks is trained per fold (25 total across the five-fold CV). Predictions at test time are the arithmetic mean of the five ensemble members, which reduces variance and improves calibration.

4.1.5. Results

Table 7 reports the neural network's five-fold CV performance in both the log/linear prediction space and in real physical units. For viscosity, the network achieves a mean R^2 of 0.638 in log-space across all compounds. When restricted to the hydrocarbon subset, performance improves to $R^2 = 0.703$, consistent with the expectation that structurally simpler hydrocarbons are easier to model than oxygenated or heteroatom-containing species. For surface tension, the network achieves a mean R^2 of 0.558 across all compounds, rising to 0.623 for hydrocarbons. Fold-to-fold variance is substantial: surface tension R^2 ranges from 0.107 to 0.868 across the five folds, reflecting the sensitivity of the model to the particular compounds in each test partition.

Property	R^2 (all)	R^2 (HC)	RMSE	Compounds
Viscosity ($\ln \eta$)	0.638	0.703	1.54	856
Surface tension (N/m)	0.558	0.623	0.009	714

Table 7: Neural network direct-prediction results: five-fold compound-level CV with five-member ensembles. Viscosity R^2 is evaluated in log-space; surface tension in physical units. RMSE is the mean across folds in the respective prediction space.

Comparing the neural network to the traditional models reveals an important asymmetry. For viscosity, the Arrhenius-parameterized XGBoost model (reconstructed $R^2 = 0.274$) operates on a smaller compound set (203 compounds with sufficient temperature coverage for Arrhenius fitting), while the neural network's direct formulation exploits the full dataset of 856 compounds. The neural network's log-space R^2 of 0.638 is therefore not directly comparable to the Arrhenius reconstruction R^2, though both indicate that fingerprints alone capture only moderate viscosity variation. For surface tension, the XGBoost Arrhenius model achieves $R^2 = 0.761$ on its 654-compound set, while the neural network reaches $R^2 = 0.558$ on the same compounds. The gap likely reflects the XGBoost model's ability to exploit nonlinear feature interactions without the aggressive dimensionality reduction imposed by the neural network's Lasso preprocessing.

4.1.6. Feature Importance Analysis

XGBoost's built-in feature-importance scores (sum of split gains across all trees) reveal which input dimensions drive the predictions. Table 8 lists the ten most important features for each property, ranked by mean importance across folds. For both viscosity and surface tension, the top features are overwhelmingly RDKit molecular descriptors rather than individual fingerprint bits. The QED drug-likeness score ranks first for both properties, followed by BCUT2D descriptors that encode eigenvalues of the modified burden matrix weighted by atomic $\log P$, partial charge, and molar refractivity. Topological indices such as Balaban's J and Kappa shape indices also appear prominently, as do partial-charge statistics and EState surface-area partitions.

Rank	**Viscosity Feature**	**Score**	**Surface Tension Feature**	**Score**
1	qed	269	qed	491
2	BCUT2D_LOGPLOW	260	BCUT2D_LOGPHI	387
3	BCUT2D_CHGHI	233	MinAbsEStateIndex	372
4	VSA_EState5	217	MolLogP	337
5	BCUT2D_LOGPHI	208	BCUT2D_MWLOW	324
6	MinEStateIndex	203	BCUT2D_LOGPLOW	316
7	BCUT2D_CHGLO	188	BCUT2D_CHGLO	311
8	BalabanJ	184	BalabanJ	304
9	BCUT2D_MRHI	175	Kappa3	278
10	FpDensityMorgan1	174	VSA_EState8	278

Table 8: Top 10 XGBoost features by split-gain importance for viscosity and surface tension. All top features are RDKit molecular descriptors; no individual Morgan or MACCS fingerprint bits appear in the top 20.

Quantitatively, the top 20 features for viscosity are all RDKit descriptors, with the highest-ranked fingerprint bit appearing beyond rank 50. The cumulative importance of all RDKit descriptor features exceeds that of all 2,214 fingerprint bits by a factor of 50–60×. This finding carries two implications. First, continuous physicochemical descriptors encode more predictive information for thermophysical properties than binary substructure indicators, which is consistent with the physical intuition that properties like viscosity depend on molecular size, shape, and polarity rather than on the presence of any single substructure. Second, the Morgan and MACCS fingerprints, despite comprising over 90% of the input dimensions, contribute marginally to the models' predictive power. This motivates the introduction of NMR spectral features in Phase 2, which provide a fundamentally different, experimentally grounded representation of molecular structure.

4.1.7. Summary and Limitations

Phase 1 establishes that molecular fingerprints and computed descriptors capture moderate predictive power for thermophysical properties: $R^2 \approx 0.64$ for viscosity in log-space and $R^2 \approx 0.56$– 0.76 for surface tension, depending on the model and evaluation protocol. The hydrocarbon subset is consistently easier to predict (R^2 improves by 0.05–0.15), reflecting both the structural regularity of hydrocarbons and their narrower property distribution. Three principal limitations motivate the subsequent phases. First, fingerprints are computed from molecular structure, which requires knowing the compound's identity. For a fuel mixture of unknown composition, this prerequisite is not met. Second, the RDKit descriptors that dominate the predictions are calculated, not measured, quantities; their accuracy degrades for unusual molecular scaffolds outside the training distribution. Third, the high fold-to-fold variance indicates that the models do not generalize robustly across chemical families, suggesting that richer molecular representations are needed to improve prediction stability.

4.2. Phase 2: NMR Spectrum Models

Phase 1 established strong baselines using molecular fingerprints derived from SMILES strings, achieving hydrocarbon R^2 values above 0.9 for viscosity with XGBoost on Morgan fingerprints. However, the Phase 1 pipeline requires the molecular structure to be known at inference time, which is not always the case for novel or proprietary fuel blends. Phase 2 pursues a fundamentally different objective: predicting thermophysical properties directly from ^{1}H NMR spectral data, without requiring SMILES strings or molecular fingerprints at inference. If successful, such a model would accept a raw NMR measurement as input and return a property estimate, eliminating the need for compound identification entirely.

4.2.1. Peak List Processing and Spectrum Generation

The NMRexp database stores ^{1}H NMR data as structured peak lists rather than raw free induction decays. Each entry is a list of tuples encoding multiplicity, J-coupling constants in Hz, an integral string (e.g., "2H"), and the upper and lower bounds of the chemical shift range in ppm. The first step in the Phase 2 pipeline parses these structured strings into a list of peak dictionaries, each containing the center chemical shift (the mean of the upper and lower bounds), the numerical integral, the multiplicity label, and a list of J-coupling values.

Converting discrete peak lists into fixed-length continuous spectra is necessary because convolutional and residual architectures require uniform-length input vectors. Each parsed peak is placed on a 1200-point grid spanning 0–12 ppm and broadened using a Lorentzian lineshape,

$$L(x) = I \cdot \frac{(\gamma/2)^2}{(x - x_0)^2 + (\gamma/2)^2} \tag{2}$$

where x_0 is the chemical shift center, I is the integral (number of hydrogens), and $\gamma = 0.01$ ppm is the linewidth parameter. For peaks with resolved J-coupling information, first-order splitting patterns are generated using Pascal's triangle coefficients. A doublet with coupling constant J places two lines separated by J/ν_0 ppm (where $\nu_0 = 400$ MHz is the assumed spectrometer frequency), with relative intensities 1:1. Triplets, quartets, and higher-order multiplets follow the standard first-order splitting rules. For complex multiplets (dd, dt, dq, and similar), successive splitting is applied: each existing line is split into a doublet by each successive coupling constant. Unresolved multiplets labeled "m" with a finite shift range are represented by Gaussian envelopes whose width matches the reported range.

After all peaks have been placed and broadened, the resulting spectrum is max-normalized to the interval $[0, 1]$, yielding a fixed-length vector of 1200 intensity values suitable for input to neural network architectures.

4.2.2. Feature Extraction from Spectra and Peak Lists

While deep learning architectures can operate directly on the 1200-point spectrum, the dataset size of approximately 500 compounds with matched NMR and property data is far too small to train convolutional or transformer models reliably. This motivates the extraction of a compact, domain-informed feature vector that reduces dimensionality while preserving chemically meaningful information.

The spectrum feature extractor computes 95 features from the normalized 1200-point spectrum. These include 60 fine-grained spectral bin features (30 bins of 0.4 ppm width, each contributing a mean and maximum intensity), 27 region-based features (integral, fractional integral, and peak count for each of nine chemically significant regions: alkyl high-field 0–1 ppm, alkyl mid 1–2 ppm, alkyl α-heteroatom 2–3 ppm, oxygenated 3–4.5 ppm, olefinic low 4.5–5.5 ppm, olefinic high 5.5–6.5 ppm, aromatic low 6.5–7.5 ppm, aromatic high 7.5–8.5 ppm, and aldehyde/downfield 8.5–12 ppm), and 8 global spectral statistics (centroid, spread, skewness, kurtosis, total peak count, normalized Shannon entropy, nonzero fraction, and maximum peak height).

However, 95 spectrum features alone proved insufficient for property prediction, particularly for viscosity. The critical issue is that max-normalization of the spectrum to $[0, 1]$ destroys absolute hydrogen count information. A molecule with 6 hydrogens and a molecule with 60 hydrogens can produce nearly identical normalized spectra if their proton environments are proportionally similar. Yet these molecules will differ dramatically in molecular weight and, consequently, in viscosity.

To address this information loss, 25 additional features are extracted directly from the parsed peak list before normalization. These peaklist features include three size proxies (total hydrogen count, number of distinct peaks, and maximum single-peak integral), nine absolute hydrogen counts per chemical shift region, seven multiplicity distribution fractions (singlet, doublet, triplet, quartet, multiplet, complex, and other), three J-coupling statistics (mean, maximum, and fraction of peaks with coupling information), and three derived ratios (alkyl hydrogen fraction, aromatic hydrogen fraction, and a CH_2/CH_3 proxy computed as the ratio of 1–2 ppm integral to the sum of 0–1 and 1–2 ppm integrals). The CH_2/CH_3 proxy is particularly informative for viscosity prediction because it correlates with chain length: longer alkyl chains have higher CH_2 content relative to terminal CH_3 groups, and longer chains produce higher viscosities through increased intermolecular van der Waals interactions.

The total hydrogen count, the most important of the 25 peaklist features, serves as a strong proxy for molecular weight. This is the single most critical piece of information that raw NMR spectra encode imperfectly: while the relative ratios of proton environments are captured faithfully by the spectrum shape, the absolute scale of the molecule is only accessible through the integral values recorded in the peak list.

Combining the 95 spectrum features with the 25 peaklist features yields a 120-dimensional feature vector per compound.

4.2.3. Model Architectures

Four model families were evaluated on the Phase 2 task. Three are deep learning architectures operating on the raw 1200-point spectrum, and one is a gradient boosted tree ensemble operating on the 120-dimensional extracted feature vector.

The 1D CNN architecture (NMR1DCNN) processes the spectrum through a sequence of convolutional layers with batch normalization, ReLU activations, and max-pooling operations. In its small configuration, designed for the available dataset size of fewer than 1000 compounds, the architecture applies three convolutional layers with 16, 32, and 64 filters (kernel sizes 7, 5, and 3 respectively), each followed by max-pooling that progressively reduces the spatial dimension from 1200 to 300 to 75 to 25 points. Global average pooling and dropout ($p = 0.5$) produce a 64-dimensional embedding. The large configuration adds a fourth convolutional layer and doubles channel widths throughout (32, 64, 128, 256 filters), yielding a 256-dimensional embedding with approximately 145,000 trainable parameters.

The 1D ResNet architecture (NMR1DResNet) introduces residual connections to enable deeper feature extraction. Each residual block consists of two convolutional layers (kernel size 3) with batch normalization and a skip connection that bypasses both layers, with a 1×1 projection applied when the channel count or spatial resolution changes. The small variant stacks four residual blocks across three stages (channels 12, 24, 48, 64) with strided convolutions for downsampling, producing a 64-dimensional embedding with approximately 18,000 parameters. The large variant uses five blocks across three stages (18, 36, 72, 144 channels), yielding a 144-dimensional embedding with approximately 190,000 parameters. Both configurations apply global average pooling and dropout before the prediction head.

The Transformer architecture (NMRTransformer) takes a fundamentally different approach by operating on the peak list directly rather than on the broadened spectrum. Each peak is encoded as a 12-dimensional feature vector containing the shift center, shift range, an 8-dimensional one-hot multiplicity encoding, and two normalized J-coupling values. A two-layer MLP embeds each peak into d-dimensional space, and learned positional encodings (maximum 50 peaks) are added to preserve ordering information. The embedded sequence is processed by a Transformer encoder with multi-head self-attention. The small variant uses $d_{\text{model}} = 32$, 4 heads, 2 layers, and a feed-forward dimension of 64. The large variant uses $d_{\text{model}} = 64$, 4 heads, 4 layers, and a feed-forward dimension of 128 with approximately 130,000 parameters. Mean pooling over non-padded positions produces the final embedding.

All three neural architectures support two prediction head configurations. The direct head concatenates the embedding with three temperature features (temperature in K, its reciprocal, and its squared reciprocal) and produces a single property value through a two-layer MLP. The Arrhenius head predicts two parameters A and B from the embedding alone, from which the temperature-dependent property is computed analytically. Both heads use Kaiming initialization and dropout regularization.

The fourth model family, XGBoost, operates on the extracted 120-dimensional feature vector (95 spectrum + 25 peaklist features) rather than on the raw spectrum. This choice is motivated by the classical machine learning principle that hand-engineered features outperform learned representations when training data is severely limited [33]. With only approximately 500 compounds, XGBoost's built-in L1/L2 regularization, column subsampling, and early stopping provide far more effective overfitting control than the implicit regularization of neural network training.

4.2.4. Results and Critical Findings

The Phase 2 comparison experiments evaluated each input representation (NMR spectrum features alone, NMR full features including peaklist features, Phase 1 fingerprints matched to the same compound subset, and a combined NMR + fingerprint representation) with both Ridge regression and XGBoost, using 5-fold cross-validation with compound-level splitting to prevent data leakage. All results are reported on the hydrocarbon subset, which provides a more meaningful evaluation than the full dataset because it excludes extreme outliers from highly polar or polymeric compounds.

Input Representation	**Model**	**Visc. HC R^2**	**ST HC R^2**	**Note**
NMR spectrum (95 feat.)	Ridge	−0.63	0.16	No size info
NMR spectrum (95 feat.)	XGBoost	−0.59	0.15	No size info
NMR full (120 feat.)	Ridge	−10.4	0.16	Unstable
NMR full (120 feat.)	XGBoost	−0.39	0.40	Improved
Phase 1 FP (matched)	Ridge	0.49	0.68	Baseline
Phase 1 FP (matched)	XGBoost	0.69	0.81	Baseline
Combined (NMR + FP)	Ridge	0.40	0.64	No gain
Combined (NMR + FP)	XGBoost	0.69	0.81	Matches FP

Table 9: Phase 2 hydrocarbon R^2 comparison across input representations and models. Viscosity values are averaged across 5 folds. Negative R^2 indicates predictions worse than the training set mean. ST denotes surface tension.

The results in Table 9 reveal several important findings. NMR spectrum features alone produce negative R^2 values for viscosity, meaning the model predictions are worse than simply predicting the mean. This is not a modeling failure but an information-theoretic limitation: the max-normalized spectrum encodes the *shape* of the proton distribution but not the *scale* of the molecule. Two alkanes with identical branching patterns but different chain lengths (e.g., decane and eicosane) produce very similar normalized spectra yet differ in viscosity by nearly an order of magnitude. For surface tension, NMR spectrum features alone achieve modest positive R^2 values (0.15–0.16), consistent with the fact that surface tension depends more on functional group polarity (which is partially encoded in spectral shape) than on molecular size.

Adding the 25 peaklist features (NMR full, 120 dimensions) improves surface tension prediction with XGBoost to $R^2 = 0.40$, confirming that the absolute hydrogen count and multiplicity distribution carry information complementary to the normalized spectrum. For viscosity, however, Ridge regression becomes unstable (mean HC $R^2 = -10.4$ due to a single catastrophic fold), while XGBoost shows modest improvement to $R^2 = -0.39$, still far below useful prediction accuracy. The NMR peaklist features provide a proxy for molecular weight through the total hydrogen count, but this proxy is noisy: molecules with the same hydrogen count can differ substantially in molecular weight depending on the number of carbon, oxygen, and nitrogen atoms, none of which are directly observable in a ^{1}H NMR spectrum.

The Phase 1 fingerprints, evaluated on the same matched compound subset, achieve HC $R^2 = 0.69$ for viscosity and $R^2 = 0.81$ for surface tension with XGBoost. These represent the ceiling against which NMR-based models are compared. Importantly, combining NMR features with fingerprints (the "combined" representation) does not improve upon fingerprints alone, indicating that the NMR features carry little information beyond what is already encoded in the Morgan/RDKit fingerprint representation when the molecular structure is known.

The three deep learning architectures (1D CNN, ResNet, and Transformer) were also evaluated on the raw spectrum and peak list inputs but consistently underperformed XGBoost on extracted features. This outcome was expected given the dataset size: with approximately 500 training compounds and 1200-dimensional input spectra, the neural architectures have far too many parameters relative to training examples, even in their small configurations. The XGBoost models on 120-dimensional extracted features benefit from aggressive regularization and require far fewer effective parameters to achieve good generalization.

4.2.5. The Molecular Weight Bottleneck

The Phase 2 results expose a fundamental information bottleneck in NMR-based property prediction. Viscosity scales approximately with M^α where $\alpha \approx 1$–2 for small molecules and can be much larger near the entanglement threshold [149]. A ^{1}H NMR spectrum encodes the proton environment distribution with high fidelity but does not directly encode molecular weight. The total hydrogen count is a weak proxy for M because the hydrogen-to-heavy-atom ratio varies substantially across chemical classes. Aromatic hydrocarbons, for instance, have far fewer hydrogens per unit molecular weight than saturated alkanes.

This bottleneck has two implications for the broader research program. First, NMR alone is unlikely to achieve competitive viscosity prediction without supplemental information about molecular size. Second, the information that NMR spectra *do* encode—functional group environments, branching patterns, aromaticity, heteroatom proximity—is precisely the kind of structural information that could complement a separate estimate of molecular weight to produce accurate property predictions. This insight motivates Phase 4′s approach of combining NMR-derived descriptors with molecular weight estimation, and it explains why the combined NMR + fingerprint representation in Phase 2 matches but does not exceed fingerprint-only performance: the fingerprints already encode molecular weight implicitly through the presence of size-dependent substructure bits.

The Phase 2 results for surface tension are more encouraging. Surface tension is governed primarily by the nature of intermolecular interactions at the liquid-vapor interface, which depends on functional group polarity and hydrogen bonding capacity more than on molecular size. The NMR full feature set achieves $R^2 = 0.40$ for surface tension even without fingerprint augmentation, suggesting that NMR spectra carry meaningful information about the intermolecular interaction potential. This finding motivates continued development of NMR-based models for properties that depend on chemical environment rather than molecular scale.

4.3. Phases 3–4: Fuel Mixture Prediction and Transfer Learning

The transition from pure-compound modeling in Phases 1 and 2 to fuel mixture prediction in Phases 3 and 4 introduces a fundamentally different challenge. Aviation fuels are complex hydrocarbon mixtures containing hundreds of individual compounds, and their NMR spectra represent the superposition of all constituent signals. No ground-truth molecular structure is available for a mixture spectrum, which means the fingerprint-based and structure-aware models developed in earlier phases cannot be applied directly. Phases 3 and 4 explore two

strategies for bridging this gap: functional group decomposition and transfer learning from a large corpus of pure-compound NMR data.

4.3.1. Phase 3: Functional Group Approach

The Phase 3 strategy follows the framework proposed by Abdul Jameel et al. [114], which maps regions of a ^{1}H NMR spectrum to nine functional group categories based on characteristic chemical shift ranges. These categories, including aliphatic CH_3, CH_2, and CH groups, naphthenic (cycloalkane) protons, olefinic protons, and several classes of aromatic protons, provide a coarse but physically meaningful decomposition of a mixture spectrum. Each functional group is assigned an integral proportional to the number of hydrogen atoms in that chemical environment, yielding a nine-dimensional feature vector that summarizes the gross composition of a fuel sample.

The Phase 3 pipeline proceeds in three steps. First, XGBoost regressors are trained on pure compounds from the Phase 2 dataset, using functional group features plus temperature as inputs and viscosity (in log Pa·s) or surface tension (in N/m) as targets. Second, identical functional group features are extracted from each fuel mixture's ^{1}H NMR spectrum. Third, the pure-compound models predict mixture properties at the temperatures for which measured data are available. Unit conversions are applied at the output stage: viscosity predictions in Pa·s are converted to kinematic viscosity in cSt using approximate fuel densities [119], and surface tension predictions in N/m are converted to mN/m.

The pipeline was validated against measured viscosity data for four AFRL reference fuels: HRJ-Camelina (POSF 7720), Shell SPK (POSF 5729), Sasol IPK (POSF 7629), and Gevo ATJ (POSF 10151). Three feature representations were compared: raw NMR spectral features (95 dimensions), full NMR features including peak-list statistics (120 dimensions), and functional groups (9 dimensions). Table 10 summarizes the per-fuel MAPE for viscosity prediction under the functional group representation, which achieved the best overall performance among the three.

Fuel	**Temps.**	**MAPE (%)**
HRJ-Camelina (7720)	233, 253 K	51.2
Shell SPK (5729)	253 K	30.6
Sasol IPK (7629)	233, 253 K	73.4
Gevo ATJ (10151)	233, 253 K	60.1
Average		**53.8**

Table 10: Phase 3 viscosity MAPE by fuel using functional group features. Measurements at low temperatures (233–253 K) are the most challenging and the most operationally relevant.

The 53.8% average MAPE is unsatisfactory for engineering use but informative as a baseline. Several factors contribute to the poor performance. The functional group representation collapses a 95-dimensional NMR spectrum into only 9 features, discarding fine structural detail that distinguishes, for example, normal alkanes from branched isomers. More critically, the training data consist entirely of pure compounds, and the model has no mechanism to account for mixture effects such as non-ideal mixing behavior or the statistical averaging inherent in a superposition spectrum. The functional group approach performs substantially better on surface tension, where the NMR spectral feature set achieves MAPEs between 1.9% (Shell SPK) and 12.9% (Jet-A), suggesting that surface tension is less sensitive to the fine structural details lost in the functional group reduction [148].

4.3.2. Phase 4: Transfer Learning Pipeline

Phase 4 introduces a two-stage transfer learning architecture designed to recover molecular-level information from NMR spectra without requiring explicit molecular structure. The core idea is to pretrain a neural network on the large NMRexp database [13], teaching it to predict molecular descriptors from NMR features alone, and then use those predicted descriptors as enriched features for viscosity regression.

4.3.2.1. The Data Bottleneck: Overlap Analysis

Before constructing the Phase 4 pipeline, it was necessary to quantify the actual volume of training data available at the intersection of NMR spectra and viscosity measurements. An overlap analysis was performed across all NMR databases (NMRexp, NMRBank, NMRShiftDB2, SDBS, NP-MRD, HMDB, BMRB, and several smaller sources) and all viscosity databases (ThermoML, the J. Cheminformatics 2024 dataset, and group-contribution predictions from the chemicals library). Each source was reduced to a set of canonical SMILES strings, and pairwise intersections were computed.

The results reveal a severe data bottleneck. The union of all NMR sources contains approximately 1.8 million unique compounds, and the union of all viscosity sources contains approximately 2,238 unique compounds. However, only 1,330 compounds appear in both collections. After filtering for valid NMR peak lists, successful feature extraction, and reasonable viscosity ranges (10^{-5} to 30 Pa·s, 200–500 K, atmospheric pressure), the final training set comprises 953 unique compounds contributing 10,633 measurement rows. Of these compounds, the large majority originate from the NMRexp database, with smaller contributions from NMRShiftDB2 and NMRBank.

This 953-compound training set is heavily biased toward polar, drug-like molecules because NMRexp draws predominantly from pharmaceutical and metabolomics literature [13]. Only a small fraction of these compounds are hydrocarbons, creating a distribution mismatch with the target application domain of aviation fuels, which are almost exclusively composed of alkanes, cycloalkanes, and aromatics.

4.3.2.2. Phase 4 v1: XGBoost Two-Stage Baseline

The Phase 4 v1 model establishes a baseline for the two-stage architecture using XGBoost [33] for both stages. In Stage 1, an XGBoost regressor is trained on the 953 viscosity compounds to predict eight RDKit molecular descriptors (molecular weight, LogP, topological polar surface area, heavy atom count, hydrogen count, rotatable bond count, aromatic ring count, and number of NMR peaks) from the 120-dimensional NMR feature vector. In Stage 2, a second XGBoost regressor takes the concatenation of NMR features (120 dimensions), predicted descriptors (8 dimensions), and temperature features (3 dimensions) as input and predicts log viscosity.

This approach improves substantially over Phase 3, reducing the average fuel MAPE from 53.8% to 39.4%. The improvement confirms that the two-stage decomposition provides useful inductive bias: by first predicting interpretable molecular properties and then using those properties alongside raw NMR features for viscosity prediction, the model gains access to structural information that is not directly accessible from the superposition spectrum of a mixture.

4.3.2.3. GPU Pretraining: NMR to Descriptors MLP

The limitation of v1 is that Stage 1 is trained on only 953 compounds, far too few to learn a robust mapping from NMR spectra to molecular descriptors. Phase 4 addresses this by pretraining a multi-layer perceptron (MLP) on the full NMRexp database. The pretraining set contains approximately 296,000 compounds for which both valid ^{1}H NMR peak lists and RDKit-computable descriptors are available.

The MLP architecture consists of three hidden layers (256, 256, 128 neurons) with batch normalization, ReLU activation, and 20% dropout after each hidden layer. The input is a 120-dimensional NMR feature vector (95 spectrum features plus 25 peak-list features), and the output is a simultaneous prediction of all eight target descriptors. Training uses the AdamW optimizer with a learning rate of 10^{-3}, cosine annealing to 10^{-5} over 30 epochs, a batch size of 4,096, and a 90/10 train/validation split. The model converges to an average validation R^2 of approximately 0.85 across the eight descriptors, indicating that NMR features carry substantial information about molecular properties even in the absence of explicit structural input.

4.3.2.4. The Distribution Shift Problem

Deploying the pretrained MLP directly on the 953 viscosity compounds (Phase 4 v2) yields an average fuel MAPE of 44.9%, which is worse than the XGBoost-only v1 baseline of 39.4%. Diagnosing the failure reveals a severe distribution shift between the pretraining domain and the target domain. The pretrained MLP was trained on 296,000 compounds dominated by heteroatom-rich pharmaceutical molecules with high TPSA values, multiple

aromatic rings, and complex substitution patterns. The viscosity training set, and aviation fuels in particular, consists largely of simple hydrocarbons with zero TPSA, zero or few aromatic rings, and high LogP values.

The mismatch manifests as physically unreasonable predictions when the pretrained MLP is applied to hydrocarbon NMR spectra. For example, the pretrained model predicts nonzero TPSA and incorrectly estimates the number of aromatic rings for pure alkanes. Evaluated on the 943 viscosity compounds before fine-tuning, the pretrained MLP achieves $R^2 = -0.99$ for molecular weight prediction, meaning its predictions are worse than simply predicting the mean value. This catastrophic failure on the most basic molecular descriptor demonstrates that the pretrained model has not learned a domain-general NMR-to-structure mapping; rather, it has learned correlations specific to the pharmaceutical chemical space of NMRexp.

4.3.2.5. Phase 4 v3: Fine-Tuned MLP (Best Result)

Phase 4 v3 resolves the distribution shift by fine-tuning the pretrained MLP on the 943 viscosity compounds before using it as a descriptor predictor. The fine-tuning procedure maintains the pretrained NMR feature scaling (since NMR spectral statistics are relatively domain-invariant) but re-fits the target scalers on the viscosity compound distribution, where descriptors like TPSA and LogP have different means and variances than in the pharmaceutical pretraining set. A differential learning rate strategy is employed: backbone layers are updated at a learning rate of 10^{-4}, while the output head receives 10^{-3}, allowing the final layer to adapt rapidly to the new target distribution without destroying the feature representations learned during pretraining.

Fine-tuning proceeds for 150 epochs with a batch size of 256, a 15% validation hold-out, and early stopping by best validation loss. After fine-tuning, the molecular weight R^2 on viscosity compounds improves from -0.99 to a positive value, and predictions for TPSA and aromatic ring count become physically reasonable for hydrocarbons (near-zero for pure alkanes).

The fine-tuned MLP descriptors, concatenated with NMR features and temperature, are then used to train the Stage 2 XGBoost viscosity model using the same architecture as v1. Table 11 shows the per-fuel viscosity MAPE across all Phase 4 versions.

Fuel	**Phase 3**	**v1**	**v2**	**v3**	**v4**
HRJ-Camelina (7720)	51.2	–	–	–	–
Shell SPK (5729)	30.6	–	–	–	–
Sasol IPK (7629)	73.4	–	–	–	–
Gevo ATJ (10151)	60.1	–	–	–	–
Average	**53.8**	**39.4**	**44.9**	**29.3**	**55.8**

Table 11: Average fuel viscosity MAPE (%) across Phase 3 and Phase 4 versions. Phase 3 uses functional groups; v1 uses XGBoost for both stages; v2 uses the raw pretrained MLP; v3 fine-tunes the MLP on viscosity compounds; v4 adds hydrocarbon and cold-temperature sample weighting. Per-fuel breakdowns for v1, v2, and v4 are not preserved in the output logs, but the averages are reported from the pipeline summaries.

The v3 pipeline achieves an average MAPE of 29.3%, reducing the error by 46% relative to the Phase 3 functional group baseline and by 26% relative to the Phase 4 v1 XGBoost-only approach. This result confirms that the pretrained MLP provides useful inductive bias, even on a very different chemical domain, once it is adapted to the target distribution through fine-tuning. The fine-tuned model effectively learns to project hydrocarbon NMR spectra into a descriptor space that captures molecular weight, chain length, and branching information relevant to viscosity.

4.3.2.6. Failed Experiments

A number of alternative approaches were explored after the v3 result, none of which improved upon the 29.3% average MAPE. These negative results are documented here because they are informative about the nature of the problem.

Hydrocarbon and cold-temperature sample weighting (v4). Because aviation fuels are predominantly hydrocarbons and the most operationally important predictions are at low temperatures (233–253 K), v4 applied

sample weights to the XGBoost training: hydrocarbon compounds received 3× weight, and cold-temperature measurements received up to 5× weight via a linear ramp from 1× at 300 K to 5× at 220 K. The expectation was that focusing the model on the relevant region of chemical and temperature space would improve fuel predictions. Instead, the average MAPE increased to 55.8%, likely because the weighting scheme reduces the effective training set size by concentrating gradient updates on a small subset of compounds while deprioritizing the diverse polar compounds that help the model learn general viscosity–structure relationships.

Arrhenius-parameterized neural network. A physics-informed model was tested that predicted the Arrhenius parameters A and B in Equation 1 directly from NMR features, rather than predicting log viscosity at each temperature independently. This approach achieved a 66% average MAPE, worse than all other variants. The failure likely stems from the sensitivity of Arrhenius extrapolation to small errors in the fitted parameters: a modest error in B translates into large viscosity errors at the extreme temperatures (233 K) where fuel predictions are validated.

MLP embedding features. Instead of using the MLP's output predictions as features, intermediate hidden-layer activations (128-dimensional embeddings from the last hidden layer) were extracted and concatenated with NMR features for viscosity prediction. This approach achieved 55% MAPE, suggesting that the hidden representations do not provide a more informative feature space than the explicit descriptor predictions.

Cold-temperature data augmentation. The training set contains only 355 measurements below 260 K out of 10,633 total rows, creating a severe cold-temperature data gap. An augmentation strategy was tested that generated synthetic low-temperature viscosity points by extrapolating Arrhenius fits for compounds with multiple temperature measurements. The augmented model achieved 58% MAPE, indicating that the extrapolated data introduced more noise than signal, which is consistent with the known breakdown of simple Arrhenius behavior near glass transition temperatures for complex molecules.

Ensemble of v3 and weighted models. An ensemble averaging the predictions of the v3 model and the v4 weighted model achieved 34% MAPE, slightly worse than v3 alone. The weighted model's large errors on specific fuels dragged down the ensemble average.

4.3.2.7. Phase 4 v5: Expanded Training Data with Synthetic NMR

A natural hypothesis for improving fuel predictions is that more training data should help, particularly if the additional compounds are hydrocarbons. The v5 experiment tested this by expanding the training set from 953 compounds (all with real NMR spectra) to 6,223 compounds (1,144 with real NMR and 5,079 with synthetic NMR). The synthetic NMR features were generated from SMILES using a substructure-based chemical shift prediction module that assigns approximate ^{1}H shifts based on standard NMR lookup tables [5]. Each hydrogen atom in the molecule is matched against a prioritized list of SMARTS patterns (aldehydes at 9.7 ppm, aromatic protons at 7.2 ppm, aliphatic CH_3 at 0.9 ppm, and so on), and the assigned shifts are broadened and binned through the same feature extraction pipeline used for real NMR data. The synthetic features are approximate, with an accuracy of roughly 0.5–1.0 ppm compared to experimental spectra, but they capture the gross spectral topology of each molecule.

The expanded dataset contains 34,460 measurement rows (compared to 10,633 in v1) and 5,442 rows below 260 K (compared to 355 in v1), substantially improving coverage at low temperatures. Three training configurations were evaluated: model A weighted real NMR data 3× higher than synthetic; model B used all data with uniform weights; and model C trained on real NMR data only (1,144 compounds, slightly more than the original 953 due to the expanded viscosity source coverage). The best of the three, model B (unweighted), achieved an average MAPE of 41.3%. All three configurations performed worse than v3′s 29.3%.

This counterintuitive result, where more data degrades performance, is explained by the noise characteristics of synthetic NMR features. The fine-tuned MLP was trained exclusively on real NMR data, and when presented with synthetic spectra at inference time, it produces descriptor predictions with systematic biases that the Stage 2 XGBoost model has not been trained to correct. The synthetic spectra lack the fine structure (coupling patterns, solvent effects, conformational averaging) that the MLP learned to exploit during pretraining, and the resulting descriptor errors compound through the two-stage pipeline. Even model C, which trains Stage 2 on real NMR data only but drawn from a slightly different compound set, fails to match v3, confirming that the specific 953-

compound training set and its tight coupling with the fine-tuned MLP represent a local optimum that is difficult to improve upon with the available data.

4.3.3. The Fundamental Bottleneck

The progression from Phase 3 (53.8% MAPE) through Phase 4 v3 (29.3% MAPE) demonstrates that transfer learning from large NMR databases can substantially improve fuel property prediction, but the final accuracy remains insufficient for engineering certification, where viscosity errors below 5–10% are typically required [23]. The fundamental bottleneck is data quality at the NMR-viscosity intersection. Out of approximately 1.8 million compounds with experimental NMR spectra, only 953 also have viscosity measurements and valid NMR features. This tiny overlap set is dominated by polar, drug-like molecules rather than the hydrocarbons that constitute aviation fuels. The pretrained MLP learns strong NMR-to-descriptor mappings on 296,000 compounds, but those mappings degrade catastrophically outside the pharmaceutical domain, and fine-tuning on 953 compounds provides only a partial correction.

Three specific gaps limit further progress. First, the NMR databases contain very few simple hydrocarbons because these compounds are of limited interest to the pharmaceutical and metabolomics communities that generate most NMR data. Second, the viscosity databases contain limited coverage below 260 K, precisely the temperature range that matters most for jet fuel pumpability at altitude. Third, the two-stage architecture introduces compounding errors: any inaccuracy in Stage 1 descriptor prediction propagates into and is amplified by Stage 2 viscosity regression. Closing these gaps would require either targeted acquisition of hydrocarbon NMR spectra with paired viscosity measurements, or the development of models that can learn directly from mixture spectra without relying on pure-compound training data as an intermediary.

5. Methodology

5.1. Evaluation Metrics

All models in this work are assessed with three complementary metrics that together characterize accuracy, scale sensitivity, and practical relevance. Let y_i denote the i-th observed value, $\hat{y}_i$ the corresponding prediction, and $\overline{y}$ the mean of the observed values over n test samples.

The coefficient of determination quantifies the fraction of variance in the observed data that is explained by the model.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \overline{y})^2} \tag{3}$$

An R^2 of 1.0 indicates perfect prediction, while a value of 0.0 indicates that the model performs no better than predicting the mean. Negative values are possible when the model is systematically worse than the mean baseline. The implementation guards against a degenerate denominator: if the total sum of squares is zero (all observations identical), R^2 is returned as zero rather than undefined.

Root mean squared error measures prediction accuracy in the same units as the target variable.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{4}$$

Because RMSE is computed after back-transformation to physical units (Pa·s for viscosity, N/m for surface tension), it provides an absolute sense of the error magnitude that can be compared directly against measurement uncertainty and specification tolerances.

Mean absolute percentage error expresses accuracy as a dimensionless fraction of the true value, making it interpretable across properties that differ by orders of magnitude.

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \tag{5}$$

To avoid division by zero, observations with $|y_i| < 10^{-10}$ are excluded from the MAPE calculation. MAPE is reported as a percentage and is the primary metric used when comparing models across properties, because a 5% error in viscosity and a 5% error in surface tension carry similar practical implications despite the values differing by several orders of magnitude.

5.2. Cross-Validation Strategy

A naive random split of individual data points would allow training and test sets to share measurements from the same compound at different temperatures. Because the temperature dependence of viscosity and surface tension is smooth and well-behaved for any single compound, such leakage would inflate apparent accuracy: the model could interpolate along a known curve rather than generalize to unseen molecules. To eliminate this source of optimistic bias, all cross-validation in this work is performed at the compound level.

Specifically, the set of unique canonical SMILES strings is partitioned into $k = 5$ folds using scikit-learn's `KFold` with a fixed random seed of 42. All temperature–property pairs belonging to a given compound are assigned to the same fold. When fold j serves as the test set, folds $1, ..., j-1, j+1, ..., 5$ provide the training data. This procedure ensures that the model has never seen any measurement from any test compound during training, providing an honest estimate of generalization to new molecules [33].

Within each training fold, a further random 90/10 split of individual data points (not compounds) is used to create a validation set for early stopping. This inner split need not be compound-level because its purpose is to detect overfitting of the current training run rather than to estimate generalization error. For XGBoost models, the validation set monitors the evaluation metric and halts boosting when performance has not improved for a specified number of rounds. For neural network models, validation loss triggers early stopping with a patience of 30–50 epochs, depending on the phase.

5.3. Hydrocarbon-Focused Evaluation

Although the training set spans a broad chemical space including alcohols, ethers, esters, and other heteroatom-containing compounds, the downstream application of this work is aviation fuel property prediction. Jet fuels are almost exclusively hydrocarbon mixtures: petroleum-derived Jet A-1 consists of n-alkanes, isoalkanes, cycloalkanes, and aromatics, while approved SAF blending components are similarly constrained to hydrocarbons or near-hydrocarbons [22], [24]. A model that achieves excellent aggregate metrics may still perform poorly on the hydrocarbon subset if it relies on heteroatom-specific features to drive its predictions.

For this reason, every results table reports two sets of metrics: one computed over all test compounds and one restricted to hydrocarbon-only (HC) compounds, defined as molecules whose atoms are exclusively carbon and hydrogen. The HC filter is applied by parsing the canonical SMILES with RDKit and checking that the set of atomic symbols is a subset of $\{\text{C}, \text{H}\}$. Hydrocarbon-only R^2 and MAPE are the primary figures of merit when evaluating suitability for fuel applications, because they reflect the chemical domain in which the model will ultimately be deployed.

5.4. Log-Viscosity as Prediction Target

Dynamic viscosity of organic liquids spans roughly six orders of magnitude across the temperature and compositional ranges encountered in this study, from approximately 10^{-5} Pa·s for light hydrocarbons near their boiling points to 10^{1} Pa·s for heavy compounds near their freezing points. Training a regression model on raw viscosity in Pa·s would cause the loss function to be dominated by the handful of high-viscosity compounds, effectively ignoring errors on low-viscosity materials that are most relevant to aviation fuels.

The Arrhenius relationship $\ln \eta = A + B/T$ provides both physical and statistical motivation for using the natural logarithm of viscosity as the prediction target [26]. In log space, viscosity is approximately linear in inverse temperature, the dynamic range compresses from six orders of magnitude to roughly 14 units, and the implicit assumption of multiplicative (rather than additive) errors aligns with the constant relative uncertainty typical of viscometric measurements. All viscosity models in this work therefore predict $\ln(\eta)$ and back-transform to Pa·s only for final metric computation and physical interpretation. Surface tension, which varies over a much narrower range (roughly 0.015–0.075 N/m), is predicted directly without transformation.

5.5. Temperature Feature Engineering

Because the target properties are temperature-dependent, every model receives temperature as an explicit input alongside the molecular representation. Rather than supplying raw temperature in Kelvin, which would place a single feature on a scale of 200–500 while fingerprint bits are binary, a three-dimensional temperature feature vector is constructed.

$$\boldsymbol{t}(T) = \left[T/1000, \quad 1/T, \quad (T/1000)^2\right] \tag{6}$$

The $1/T$ term directly encodes the Arrhenius-like inverse-temperature dependence of viscosity. The scaled temperature $T/1000$ provides a normalized linear term, and its square $(T/1000)^2$ captures any residual curvature beyond the Arrhenius approximation. This feature set allows even a linear model to represent the dominant $1/T$ dependence while giving nonlinear models (neural networks, gradient-boosted trees) additional flexibility to capture departures from ideal Arrhenius behavior. The scaling by 1000 ensures that all three features are of order unity, avoiding numerical issues during gradient-based optimization.

5.6. Data Filtering and Quality Control

Raw thermophysical data aggregated from ThermoML [30] and the chemicals library [32] undergo several filtering steps before model training. Temperature is restricted to the range 200–500 K, which covers the service envelope of aviation fuels with a margin for extrapolation studies. Pressure is limited to atmospheric or near-atmospheric conditions ($\leq$ 110 kPa) where available, because the models in this work do not include pressure as an input variable. Viscosity values are retained only if they fall within 10^{-5} to 30 Pa·s; values outside this range are flagged as likely unit-conversion errors or non-Newtonian materials and are discarded. Surface tension values are similarly bounded between 0 and 1 N/m.

Molecular features are computed using RDKit [147]. Morgan fingerprints (radius 2, 2048 bits) and a suite of RDKit molecular descriptors are generated for each compound from its canonical SMILES. Compounds that yield `NaN` or infinite descriptor values after sanitization are removed. Fingerprint bits and descriptors with zero variance across the training set are dropped, because they carry no discriminative information. When ThermoML and group-contribution sources provide overlapping measurements for the same compound at the same temperature (rounded to 0.1 K), the ThermoML value is preferred on the grounds that it originates from peer-reviewed experimental data.

SMILES strings are canonicalized through RDKit's `MolFromSmiles` and `MolToSmiles` round-trip to ensure consistent molecular representation. InChI identifiers from ThermoML are first converted to RDKit molecule objects via `MolFromInchi` and then output as canonical SMILES via `MolToSmiles`. This two-step canonicalization guarantees that the same compound is represented identically regardless of its original identifier format, preventing duplicate entries and ensuring correct feature–label joins.

5.7. Training and Early Stopping

Neural network models are trained with the AdamW optimizer at an initial learning rate of 3×10^{-4} and weight decay of 10^{-4}. A cosine annealing schedule with 10-epoch linear warm-up reduces the learning rate to a floor of 10^{-6} over the course of training. Gradient norms are clipped to 1.0 to prevent instability. The loss function is the Huber loss with $\delta = 1.0$, which combines the sensitivity of mean squared error near zero with the robustness of mean absolute error for large residuals, providing resilience against occasional outlier measurements in the

aggregated dataset. Training proceeds for up to 500 epochs with early stopping triggered after 50 epochs of no improvement on the validation set.

For each cross-validation fold, an ensemble of five independently initialized models is trained with different random seeds. At inference time, the ensemble prediction is the arithmetic mean of the five individual predictions. Ensembling reduces variance without increasing bias and provides a simple mechanism for uncertainty quantification through the spread of individual predictions.

XGBoost models [33] use an analogous early-stopping protocol. The validation set monitors the evaluation metric (RMSE in log space for viscosity, RMSE in original units for surface tension), and boosting terminates when the metric has not improved for a specified number of rounds. Hyperparameters including maximum tree depth, learning rate, subsample ratio, and regularization terms are selected via grid search within each cross-validation fold.

Feature selection is performed via Lasso regression (ℓ_1-penalized linear model) applied to the compound-level mean of each target property. Features whose Lasso coefficients exceed 10^{-6} in absolute value for any target are retained; all others are discarded. This step reduces the input dimensionality from several thousand (2048 fingerprint bits plus several hundred RDKit descriptors) to a few hundred informative features, improving training speed and reducing overfitting risk. If Lasso selects fewer than 10 features (indicating an overly aggressive penalty), the full feature set is retained as a fallback.

5.8. Hardware and Software

All experiments are conducted on a shared compute server (comech-2422) equipped with an NVIDIA RTX 6000 Ada Generation GPU (48 GB VRAM). PyTorch models are trained on the GPU with mixed-precision arithmetic where supported. XGBoost training uses CPU-based histogram methods, which complete within minutes for the dataset sizes in this study.

The software stack consists of Python 3.10 with the following principal libraries: RDKit [147] for molecular parsing, fingerprint generation, and descriptor computation; XGBoost [33] for gradient-boosted tree models; PyTorch for neural network construction and training; and scikit-learn for preprocessing (standard scaling), cross-validation splitting, and Lasso feature selection. All random seeds are fixed at 42 (or deterministic offsets thereof) to ensure reproducibility of data splits, model initialization, and training dynamics.

6. Discussion

The results across all four phases of this work converge on several conclusions about the strengths, limitations, and future directions of NMR-based thermophysical property prediction. This section interprets the key findings, examines the failure modes encountered, and identifies the practical requirements for deploying NMR-based models in a fuel qualification context.

6.1. What NMR Spectra Encode and What They Miss

The most consistent finding across all experiments is the asymmetry between surface tension and viscosity prediction. Surface tension models achieved hydrocarbon R^2 values exceeding 0.81 with NMR features combined with molecular descriptors in Phase 2, and fuel-level surface tension predictions reached MAPE values as low as 1.9% for Shell-SPK and 3.0% for Gevo-ATJ using the NMR spectrum representation in Phase 3. Viscosity, by contrast, proved far more difficult. The best pure-compound viscosity model in Phase 1 achieved a mean hydrocarbon R^2 of only 0.70 across five folds even with molecular fingerprints that encode molecular weight, branching, and ring topology. NMR spectra alone performed substantially worse, with the Phase 2 NMR-only XGBoost model yielding negative R^2 values on multiple viscosity folds.

This disparity has a clear physical explanation. Surface tension is dominated by intermolecular interactions at the liquid–air interface and scales approximately linearly with temperature for hydrocarbons. The functional group

inventory encoded in a ^{1}H NMR spectrum, which directly quantifies aromatic, olefinic, and aliphatic content, captures the essential determinants of these interfacial forces [21]. Viscosity, however, depends on the resistance to molecular rearrangement within the bulk liquid, which is exponentially sensitive to temperature through the Arrhenius relationship $\ln \eta = A + B/T$. The pre-exponential factor A and the activation energy parameter B both depend strongly on molecular weight, chain length, and branching patterns that NMR chemical shifts encode only indirectly. A methylene peak at 1.3 ppm reports the presence of CH_2 groups but does not distinguish a C_{10} chain from a C_{20} chain. Because viscosity scales roughly as a power law with molecular weight [3], this ambiguity is catastrophic for viscosity prediction.

The combined NMR-plus-fingerprint representation in Phase 2 achieved HC $R^2 = 0.694$ for viscosity and $R^2 = 0.813$ for surface tension with XGBoost, matching but not exceeding the fingerprint-only baseline. This parity indicates that when molecular structure is known (as it must be to compute fingerprints), NMR features contribute no additional information beyond what the fingerprints already encode. The practical value of NMR emerges only in the mixture regime where molecular structure is unknown, motivating the transition to Phases 3 and 4.

6.2. Transfer Learning: Promise and Distribution Shift

The Phase 4 transfer learning experiments tested whether pretraining on a large corpus of NMR–descriptor pairs could compensate for the small number of compounds with both NMR spectra and measured viscosity. The pretraining dataset contained 296,000 compounds from the NMRexp database [13], for which molecular descriptors were computed from SMILES structures. A multilayer perceptron trained on this corpus learned a general mapping from NMR features to molecular descriptors, which could then serve as a feature extractor for downstream viscosity prediction.

The direct application of the pretrained MLP to viscosity compounds failed, producing an average fuel MAPE of 44.9% in Phase 4 v2, worse than the 39.4% achieved by the simpler Phase 4 v1 XGBoost pipeline. The failure had a straightforward cause. The NMRexp database is dominated by drug-like molecules containing nitrogen, oxygen, sulfur, and halogen substituents, while the viscosity training set consists almost exclusively of hydrocarbons containing only carbon and hydrogen. The descriptor distributions learned during pretraining reflected this drug-like bias, and the model produced systematically incorrect descriptor estimates when applied to hydrocarbon inputs. The pretrained model predicted nonzero topological polar surface area for pure alkanes and achieved $R^2 = -0.99$ for molecular weight prediction on the viscosity compounds, meaning its predictions were worse than simply predicting the mean. This is a canonical example of distribution shift in transfer learning: the source domain (drug-like organic molecules) and target domain (hydrocarbon fuels) share the same input modality (NMR features) but occupy different regions of chemical space.

Fine-tuning the pretrained MLP on the 943 viscosity compounds in Phase 4 v3 reduced the average fuel MAPE from 39.4% to 29.3%, a 26% relative improvement over the non-pretrained baseline. This result demonstrates that transfer learning can extract useful representations from large NMR databases even when the domain mismatch is severe, provided that a fine-tuning step adapts the learned features to the target chemical space. The pretrained weights gave the model an informed initialization that captured general NMR–structure correlations, and fine-tuning on the small but domain-relevant viscosity dataset refined those weights to the hydrocarbon regime.

The improvement is meaningful but bounded by the fundamental scarcity of hydrocarbon–NMR–viscosity intersections in existing databases. The 943 compounds with both NMR spectra and viscosity measurements span a narrow range of hydrocarbon classes, predominantly normal alkanes, branched alkanes, and single-ring aromatics. Polycyclic naphthenes, multi-ring aromatics, and the heterogeneous mixture of structural types found in real jet fuels remain underrepresented.

6.3. The Data Bottleneck

The most consequential finding of this work is not any individual model result but the severity of the data bottleneck at the intersection of NMR spectra and thermophysical measurements. The union of all NMR sources surveyed in Section 3.1 contains approximately 1.8 million unique compounds, and the union of all viscosity sources

in Section 3.2 contains approximately 2,238 unique compounds. The intersection is 1,330 compounds, which after filtering for valid NMR features and reasonable property ranges yields only 953 training compounds contributing 10,633 measurement rows. Of these, only 355 measurements fall below 260 K, precisely the temperature range that matters most for jet fuel pumpability at altitude.

This bottleneck is not merely a matter of scale but of distribution. The 953 overlap compounds are dominated by polar, drug-like molecules from pharmaceutical and metabolomics literature, because these fields generate the majority of NMR data deposited in public repositories. Simple hydrocarbons, which constitute the entirety of aviation fuel composition, represent only a small fraction of the overlap set. This distributional mismatch creates an extrapolation problem that no amount of architectural sophistication can fully overcome: a model trained predominantly on oxygenated and nitrogen-containing aromatics will struggle to predict properties of saturated alkanes and cycloalkanes, regardless of how expressive its functional form.

Three specific gaps are responsible. First, NMR databases contain few simple hydrocarbons because these compounds hold limited interest for the pharmaceutical and metabolomics communities that generate most public NMR data. NMRexp, the largest open NMR database at 1.48 million compounds, draws from supporting information PDFs in the biomedical literature, where hydrocarbons are underrepresented [13]. Second, viscosity databases have limited coverage below 260 K because standard viscometric methods become increasingly difficult at cryogenic temperatures. The ThermoML archive, while containing over 44,000 pure-compound viscosity points, concentrates most of its data between 273 K and 373 K [30]. Third, the two-stage architecture used in Phase 4 introduces compounding errors: any inaccuracy in Stage 1 descriptor prediction propagates into and is amplified by Stage 2 viscosity regression, a problem that becomes more severe when the Stage 1 model operates outside its training distribution.

6.4. The Cold-Temperature Extrapolation Problem

The ASTM D1655 specification sets a maximum viscosity of 8.0 mm^2/s at $-20°C$ (253.15 K) for Jet A and at $-40°C$ (233.15 K) for Jet A-1 [23]. Predicting viscosity at these cold temperatures is therefore the most practically relevant capability for fuel screening, and it is precisely where the models perform worst. Phase 3 fuel predictions at 233 K showed dramatic underestimates across all NMR representations. For Gevo-ATJ (POSF 10151), the measured viscosity at 233 K was 10.5 cSt while the NMR spectrum model predicted 0.87 cSt, an error exceeding 90%. For HRJ-Camelina (POSF 7720), the measured value was 11.4 cSt versus a prediction of 1.06 cSt. In every case the model catastrophically underestimated the viscosity at the specification-critical temperature while performing reasonably at 253 K, where predictions fell within 10–50% of measured values.

The root cause is data scarcity at low temperatures compounded by the nonlinear nature of the Arrhenius extrapolation. The training set contains only 14 compounds with viscosity measurements below 240 K. The Arrhenius parameterization $\ln \eta = A + B/T$ amplifies prediction errors at low temperatures because the B/T term grows as T decreases. A 10% error in the estimated B parameter produces a 10% error in $\ln \eta$ at 300 K but a 13% error at 233 K, and because viscosity is the exponential of $\ln \eta$, these errors compound multiplicatively in real units. The cold-temperature problem is not unique to NMR-based models. Any data-driven approach trained predominantly on measurements between 280 and 370 K will fail to extrapolate the Arrhenius curve into the 220–250 K regime. The distinction between interpolation and extrapolation is fundamental and cannot be resolved by improving the molecular representation or the model architecture alone. Resolving this limitation requires either collecting substantially more cold-temperature viscosity data for NMR-characterized hydrocarbons or incorporating physics-based constraints that enforce the correct asymptotic behavior at low temperatures.

6.5. Pure Compounds versus Fuel Mixtures

All models developed in Phases 1 through 4 were trained on pure-compound data and applied to fuel mixtures by treating the mixture NMR spectrum as if it arose from a single compound. This approximation is physically unjustified. A jet fuel typically contains hundreds of individual hydrocarbon species spanning a molecular weight range from roughly C_8 to C_{16}, with each species contributing to the bulk NMR spectrum in proportion to its mole fraction [22], [25]. The measured NMR spectrum of a fuel is the mole-fraction-weighted superposition of hundreds of individual spectra [16], [17].

When a model trained on pure compounds receives this superposition as input, it interprets the composite functional group distribution as belonging to a single hypothetical molecule. For surface tension, which depends primarily on the functional group inventory and varies modestly with molecular weight, this single-compound approximation works surprisingly well. The Phase 3 results confirm this: surface tension predictions for five reference fuels achieved MAPE values between 1.9% (Shell-SPK) and 12.9% (Jet-A POSF 10325), with the best results for Fischer-Tropsch and hydroprocessed fuels that have relatively narrow compositional distributions. These SPK-type fuels are predominantly isoparaffinic, so their NMR spectra more closely resemble those of individual compounds than do the spectra of conventional petroleum-derived fuels.

For viscosity, the mixture approximation is far more damaging. Viscosity is a nonlinear function of composition that depends on intermolecular interactions between dissimilar species. The viscosity of a binary mixture of *n*-octane and *n*-hexadecane, for example, is not the mole-fraction average of the two pure-component viscosities. Well-established mixing rules exist [3], but they require knowing the identities and concentrations of the individual components, information that is lost when the model receives only the composite NMR spectrum. The consistently high MAPE values for fuel viscosity prediction, even after transfer learning, likely reflect this fundamental mismatch between pure-compound training and mixture-level inference at least as much as they reflect shortcomings in the NMR features themselves.

6.6. Comparison to Prior Work

The results of this work can be situated within the broader landscape of NMR-based fuel property prediction. Abdul Jameel et al. demonstrated that NMR functional group features can predict octane numbers and ignition quality of gasoline-range fuels with MAPEs below 5% [19], [20]. Parker et al. extended this approach to density and surface tension for aviation fuels, achieving strong results using NMR atom type descriptors [21]. The present work's Phase 3 surface tension results are consistent with these precedents, with MAPEs between 1.9% and 12.9%, confirming that NMR functional group decomposition captures sufficient information for properties governed by intermolecular interactions at the liquid-vapor interface.

The viscosity results, however, reveal a fundamental difference between viscosity and the properties studied in prior NMR-fuel work. Octane number, cetane number, and even surface tension depend primarily on the types of chemical bonds present in the fuel, information that is directly encoded in the chemical shift distribution. Viscosity, by contrast, depends strongly on molecular weight, chain length, and intermolecular entanglement, properties that are only indirectly reflected in the ^{1}H NMR spectrum. The molecular weight bottleneck identified in Phase 2 has not been explicitly noted in prior NMR-fuel studies, likely because those studies focused on properties for which molecular size is a secondary factor.

For pure-compound viscosity prediction from molecular structure, recent machine learning approaches using graph neural networks have achieved R^2 values exceeding 0.90 on curated datasets [149]. The Phase 1 baseline of $R^2 = 0.638$ falls below this ceiling, reflecting both the heterogeneity of the training data (which spans ionic liquids to light hydrocarbons) and the use of fixed fingerprints rather than learned molecular representations. The Phase 2 combined model ($R^2 = 0.694$ for HC viscosity) narrows the gap but does not close it, consistent with the finding that NMR features provide complementary rather than superior information when molecular structure is known.

6.7. Implications for Fuel Screening

Despite the 29.3% average MAPE achieved by the best Phase 4 model, the current pipeline does not meet the accuracy requirements for fuel certification, where viscosity errors below 5–10% are typically required to ensure compliance with ASTM D1655 specifications [23]. However, the pipeline may still be useful as a first-pass screening tool in the early stages of SAF qualification. In the Tier 1 screening phase of the ASTM D4054 evaluation process, the goal is to identify candidate fuels that are clearly unsuitable, not to provide specification-grade property estimates [29]. A model that predicts viscosity to within 30% could reliably flag fuels whose viscosity is more than a factor of two outside specification limits, eliminating obviously unsuitable candidates and reserving expensive laboratory testing for the most promising formulations.

The practical workflow would proceed as follows. A candidate SAF sample undergoes a single ^{1}H NMR measurement requiring fewer than five minutes and microliters of material. The spectrum is processed through the feature extraction pipeline to produce a 120-dimensional descriptor vector. The fine-tuned MLP predicts molecular descriptors from these NMR features, and the Stage 2 XGBoost model predicts viscosity across the temperature range of interest. Candidates whose predicted viscosity curves fall within a conservative tolerance band around the specification limit proceed to physical testing; those outside are deprioritized. This workflow compresses the initial screening from days of laboratory viscometry to minutes of computational prediction.

The value of NMR as a screening measurement extends beyond prediction accuracy. An NMR spectrum is acquired non-destructively, requires minimal sample preparation, and produces a permanent digital record that can be compared against reference databases, reanalyzed as models improve, and shared across laboratories in standardized formats [126], [128]. These operational advantages mean that NMR-based screening could be integrated into the earliest stages of SAF development, where hundreds of candidate blends may need to be evaluated rapidly and only a handful will proceed to full ASTM D4054 testing.

6.8. What Would Improve Results

The analysis across all four phases points to three specific investments that would most substantially improve NMR-based fuel property prediction.

First, the single most impactful intervention would be collecting experimental viscosity measurements at temperatures between 220 and 260 K for a targeted set of hydrocarbons that already have NMR spectra in existing databases. The current training set contains only 14 compounds with data below 240 K, which is wholly insufficient to learn the steep viscosity upturn that governs specification compliance. A focused measurement campaign covering 50 to 100 representative hydrocarbons, spanning normal alkanes, isoparaffins, cycloparaffins, and mono- and polycyclic aromatics, would fill the gap that currently makes cold-temperature prediction impossible. The compounds should be selected to maximize coverage of the hydrocarbon structural space relevant to jet fuels while prioritizing species already present in HMDB [11], NMRexp [13], or SDBS [15] so that high-quality experimental NMR spectra are already available.

Second, collecting NMR spectra of fuel mixtures with simultaneously measured viscosity and surface tension would directly address the pure-compound-to-mixture transfer problem. Rather than training on pure compounds and hoping that the model generalizes to superposed spectra, a model trained on mixture NMR spectra paired with mixture property measurements would learn the correct composition–property relationship without requiring deconvolution of individual components. The NJFCP reference fuel set [22], [25] provides a natural starting point, as these fuels have extensive property characterization and several already have published NMR spectra. Expanding this dataset to include SAF blends at various blend ratios would create a training corpus directly relevant to the fuel qualification task.

Third, developing mixture-aware model architectures that explicitly account for spectral superposition would address the conceptual mismatch between pure-compound training and mixture inference. One approach would decompose the mixture spectrum into component contributions using NMR deconvolution methods, estimate pure-component properties for each identified species, and combine them using established mixing rules. An alternative would train end-to-end on mixture spectra, bypassing the deconvolution step entirely but requiring a sufficiently large training corpus of mixture NMR spectra paired with measured properties. Both approaches represent a substantial advance over the current single-compound approximation and would bring the modeling framework into alignment with the physical reality of multi-component fuel systems.

7. Conclusion

This report presented a four-phase machine-learning pipeline for predicting thermophysical properties of organic compounds and aviation fuel mixtures directly from nuclear magnetic resonance spectra. The pipeline was developed incrementally, with each phase addressing a specific limitation revealed by the preceding one, and its progression illustrates both the promise and the present constraints of NMR-based property prediction.

Phase 1 established baseline models using molecular fingerprints (Morgan, MACCS, and RDKit) as the sole input representation. XGBoost regression on these fingerprints achieved $R^2 = 0.638$ for viscosity prediction on pure compounds, confirming that structural descriptors derived from SMILES strings carry substantial information about bulk transport properties but leave a meaningful fraction of the variance unexplained. Phase 2 introduced NMR spectral features, both binned ^{1}H and ^{13}C spectra and summary statistics extracted from them. NMR features alone proved insufficient for accurate viscosity prediction, a finding attributable to the ambiguity that arises when many structurally distinct molecules share similar spectral envelopes. However, concatenating NMR descriptors with molecular fingerprints raised the best pure-compound viscosity model to $R^2 = 0.694$ and the surface-tension model to $R^2 = 0.813$, demonstrating that NMR features capture complementary physicochemical information that fingerprints miss.

Phase 3 extended prediction from pure compounds to real aviation fuel mixtures by constructing composition-weighted spectral descriptors from functional-group analyses. Applied to viscosity prediction across five POSF reference fuels spanning -20 to $80°C$, the Phase 3 model achieved 53.8% mean absolute percentage error. Although this error is too large for final fuel qualification, it represents a meaningful first demonstration of NMR-to-mixture-property prediction in a regime where training data are extremely scarce. Phase 4 attacked the data scarcity problem through transfer learning. A multilayer perceptron was first pretrained on 296,000 pure-compound NMR–viscosity pairs to learn general spectral-to-viscosity mappings and then fine-tuned on only 943 fuel-relevant compounds. This strategy reduced mixture viscosity MAPE to 29.3%, a 46% relative improvement over the Phase 3 baseline, confirming that representations learned from abundant pure-compound data transfer usefully to the mixture domain.

A critical enabler of the transfer-learning approach was the assembly of the largest known NMR feature dataset, encompassing 1.8 million compounds drawn from 14 public and proprietary spectral databases. Despite this scale, the binding constraint on model performance remains the narrow intersection between NMR coverage and experimental thermophysical measurements. Only 953 compounds currently reside at the NMR–viscosity intersection with reliable experimental data, and this bottleneck limits both the diversity of chemical space the models can learn from and the statistical power of downstream evaluations.

Several directions for future work follow naturally from these findings. First, expanding the experimental viscosity and surface-tension corpus, particularly for oxygenated and cyclic compounds prevalent in SAF candidate streams, would directly widen the training distribution and reduce extrapolation error. Second, developing mixture-aware architectures that explicitly model intermolecular interactions, rather than relying on composition-weighted averaging of pure-component descriptors, could capture nonideal mixing effects that the current pipeline neglects. Third, extending temperature coverage below $-40°C$ is essential for predicting cold-flow behavior under the most demanding flight conditions. Taken together, these advances would move NMR-based property prediction from a screening tool toward a quantitative surrogate for laboratory measurement, accelerating the evaluation and qualification of sustainable aviation fuels at a pace commensurate with the industry's decarbonization goals.

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